Implementing a Bike-Sharing System in Jordan: A Data-Driven Approach Using Insights from Capital Bikeshare

Graduation Project

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Submitted by

Dania Saleh

Waseem Mahmoud

Under the Supervision of

Dr. Abdelaziz Saleh Mohammad

Assistant Professor

Department of Business Intelligence
Business Intelligence Program
Middle East University
Amman- The Hashemite Kingdom of Jordan

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Abstract

This study investigates the usage patterns and factors influencing the performance of the Capital Bikeshare system, focusing on data from 2019 to 2024. The analysis aims to identify trends in daily and seasonal bike rentals, the impact of weather conditions, and user behavior. Key findings highlight peak usage during commuting hours and warmer seasons, with significant variations in rentals based on temperature and day of the week. The study also identifies growth opportunities, such as expanding bike station coverage, integrating with public transportation, and enhancing user experience through weather-resilient infrastructure and digital tools.

Additionally, the research explores the feasibility of implementing a similar bike-sharing system in Jordan, addressing its unique transportation challenges, urban landscape, and climate. Recommendations include developing shaded bike lanes, providing low-cost rental options, and promoting cycling as a healthy and sustainable mode of transport. The study emphasizes the importance of tailored infrastructure, community engagement, and sustainable practices to ensure the success of such systems. These findings provide actionable insights for policymakers, urban planners, and transportation service providers to enhance the efficiency and accessibility of bike-sharing systems in diverse contexts.

Keywords: Bike-sharing systems, Capital Bikeshare, Usage patterns, Seasonal trends, Weather impact, Urban mobility, Sustainable transportation, Cycling infrastructure, User behavior, Jordan bike-sharing feasibility

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Table of Contents

Abstract 2

Acknowledgement 3

List of Figures 8

Introduction 12

Problem Statement 13

Project Scope 14

Objectives of the Study 15

Overview of Capital Bikeshare 16

Literature Review 19

Temporal Trends in Bike Rentals 19

Weather's Impact on Usage Patterns 20

Popularity of Specific Stations 20

Recommendations for Service Improvement 21

Broader Implications for Urban Transportation 21

Promoting Physical Activity 22

Environmental Benefits 22

Enhancing Urban Mobility 23

Infrastructure Development 23

Conclusion 25

Methodology 27

جامعة الشرق الأوسط
MIDDLE EAST UNIVERSITY

Data Collection 27

Data Sources 27

Data Cleaning and Preprocessing 29

Comprehensive Data Analysis 29

Key Research Questions 29

Time-Series Forecasting 30

Interactive Dashboard 30

Recommendations for Implementing a Bike-Sharing System in Jordan 30

Technical Requirements 31

Data Analysis 32

Data Cleaning & Preprocessing 32

Combining the Data 32

Inspecting the Data 37

Data Cleaning 42

Linking with Weather Data 45

Imputing Missing Values 46

Detecting Outliers & Finalizing the Data 49

Comprehensive Data Analysis 53

Preparing for Data Analysis 53

Quarterly Rentals per Year 55

MEU
جامعة الشرق الأوسط
MIDDLE EAST UNIVERSITY

Yearly Rental Trends 59

Daily Rental Trends 60

Hourly Rental Trends 64

Top 10 Popular Stations 67

Distribution of Trip Durations by User Type 68

Average Trip Durations by Hour for each User Type 69

Weather Correlation 70

Time-Series Forecasting 73

Interactive Dashboard 76

Results & Findings 79

Discussion 81

Temporal and Seasonal Trends 81

Station Popularity 81

Impact of Weather 81

User Behavior 82

Recommendations for System Improvement 82

Recommendations for Implementing a Bike-Sharing System in Jordan 83

Conclusion 84

References 86

Appendix 90

MEU
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MIDDLE EAST UNIVERSITY

Appendix A: Raw Data Description 90

Appendix B: Data Tables 91

Appendix C: Limitations of the Study 92

MEU
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MIDDLE EAST UNIVERSITY

List of Figures

Code 1. Importing the Data 32

Code 2. Mapping the Data 33

Code 3. Initial Cleaning 34

Code 4. Combining Dataframes 34

Code 5. Saving & Loading 34

Code 6. General Sampling 36

Code 7: Stratified Sampling 37

Code 8: Data Inspection using head(), summary(), and str() 38

Code 9. Removing Unnecessary Columns 42

Code 10. Converting date-time data into POSIXct 43

Code 11. Checking for Duplicate Rows 43

Code 12. Removing Observations with Missing Values 44

Code 13. Standardizing the Data 44

Code 14. Linking with Weather Data 46

Code 15. Imputing Data using Mean 47

Code 16. Imputing Data using Mode 47

Code 17. Imputing Data using Arithmetic Operations 48

Code 18. Visualizing Distributions 49

Code 19. Creating the IQR Outlier Detection Function 51

Code 20. Applying the Function 51

Code 21. Detecting Outliers 51

Code 22. Handling the Outliers 52

MEU
جامعة الشرق الأوسط
MIDDLE EAST UNIVERSITY

Code 23. Using summary() on Clean Data 53

Code 24. Quarterly Rentals per Year 56

Code 25. Yearly Rental Trends 59

Code 26. Daily Rental Trends 61

Code 27. Hourly Rental Trends 65

Code 28. Top 10 Popular Stations 68

Code 29. Distribution of Trip Durations by User Type 68

Code 30. Average Trip Durations by Hour for Each User Type 69

Code 31. Weather Correlation 72

Code 32. Time-Series Forecasting 75

Figure 1. Capital Bikeshare Membership Pricing 17

Figure 2. Capital Bikeshare System Data 28

Figure 3. Washington, D.C. Weather Data 28

Figure 4. Using str() to Inspect the Data 39

Figure 5. Using summary() to Inspect the Data 41

Figure 6. Visualizing Missing Values 42

Figure 7. Temperature Boxplot 49

Figure 8. Precipitation Boxplot 50

Figure 9. Precipitation Boxplot After Removing Outliers 52

Figure 10. Temperature Boxplot After Removing Outliers 52

Figure 11. Using summary() on Cleaned Data 54

Figure 12. Quarterly Rentals for 2019 57

MEU
جامعة الشرق الأوسط
MIDDLE EAST UNIVERSITY

Figure 13. Quarterly Rentals for 2020 57
Figure 14. Quarterly Rentals for 2021 57
Figure 15. Quarterly Rentals for 2022 58
Figure 16. Quarterly Rentals for 2023 58
Figure 17. Quarterly Rentals for 2024 58
Figure 18. Yearly Rentals from 2019 to 2024 60
Figure 19. Daily Rentals for 2019 62
Figure 20. Daily Rentals for 2020 62
Figure 21. Daily Rentals for 2021 62
Figure 22. Daily Rentals for 2022 63
Figure 23. Daily Rentals for 2023 63
Figure 24. Daily Rentals for 2024 63
Figure 25. Hourly Rentals for 2019 65
Figure 26. Hourly Rentals for 2020 65
Figure 27. Hourly Rentals for 2021 66
Figure 28. Hourly Rentals for 2022 66
Figure 29. Hourly Rentals for 2023 66
Figure 30. Hourly Rentals for 2024 67
Figure 31. Top 10 Popular Stations 68
Figure 32. Distribution of Trip Durations 69
Figure 33. Trip Durations by User Type 69
Figure 34. Average Trip Duration by Hour 70
Figure 35. Temperature vs. Average Duration 72

MEU
جامعة الشرق الأوسط
MIDDLE EAST UNIVERSITY

Figure 36. Temperature vs. Total Rentals 72

Figure 37. Forecast of Bike Rentals Across Multiple Years 75

Figure 38. Tableau Dashboard Page One 76

Figure 39. Tableau Dashboard Page Two 76

Figure 40. Power BI Dashboard 77

Snapshot 1. Combining the Data 35

Snapshot 2. General Sample 36

Snapshot 3. Stratified Sampling 37

Snapshot 4. Data Cleaning 45

Snapshot 5. Merging the Data 46

Snapshot 6. Imputing the Data 48

Snapshot 7. Detecting & Handling the Outliers 53

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Introduction

The Capital Bikeshare program, launched in Washington, D.C., in 2010, has emerged as a cornerstone of urban mobility and sustainable transportation. The initiative was made to mitigate traffic jams and intervene in reducing the carbon footprint of commuting. This bike-sharing initiative grew by leaps and bounds gradually, now extending to thousands of bicycles and over 600 stations open to the public (Capital Bikeshare, n.d.), filling an obvious void by providing seamless services while campaigning for cycles as much better lifestyle choices.

Primarily, the objectives of the program are to avail the means of moving people in and around the city seamlessly and to encourage even casual people to cycle as a healthier form of lifestyle. As congestion and pollution continue to plague most urban areas, it seems bike-sharing systems such as Capital Bikeshare offer a realistic remedy (Fishman, Washington, & Haworth, 2014). Cities like Paris have already executed such programs that benefited from changes in the conditions of urban transport (Shaheen & Guzman, 2011).

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Additionally, cycling contributes to the conservation of the environment by eliminating car trips, most especially the short ones, thus leading to less emission of greenhouse gases and improved air quality (Politis, Fyrogenis, Papadopoulos, Nikolaidou, & Verani, 2020). Economically, bike-sharing systems lower travel costs for users and can reduce healthcare costs associated with lifestyle-related illnesses.

Users pay a small fee just to rent a bicycle for a while, making it easy to integrate cycling with public transit. The success of Capital Bikeshare has led many cities in the USA and around the world to consider similar developments and seriously look at cycling as a transport mode in urban areas.

Despite such an advantage, there are additional countries besides Jordan, such as those that are not limited to transport system inefficiencies but are also experiencing increasing health crises due to the sedentary lifestyle afflictions of the modern person (Park, Moon, Kim, Kong, & Oh, 2020). Limited options in public transport inhibit individuals from engaging in even a slight amount of activity that would contribute to the public health concern, although an increase in public transport usage among inactive adults could lead to a significant rise in the overall population of sufficiently active adults (Rissel, Curac, Greenaway, & Bauman, 2012), highlighting the need for implementing a bike-sharing system.

This study is intended to discuss usage patterns in Capital Bikeshare between 2019 and 2024 relevant to insights derived from usage patterns to enhance operational efficiency and user satisfaction. The study would bring up the results from this successful case of bike-sharing and provide recommendations for a similar implementation in Jordan to bring improvements to various aspects.

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Problem Statement

Notwithstanding the growing attention towards bike-sharing systems as immediate strategies for urban mobility and public health, Jordan faces serious problems in effective transport systems (Shbeed, 2018), environmental sustainability (Hadadin & Tarawneh, 2007), and increasing health crises linked to issues such as obesity and cardiovascular diseases (Ajlouni, Khader, Batieha, Jaddou, & El-Khateeb, 2020). If people have limited options for accessible and convenient transportation, it will limit their potential for engaging in physical activities, hence worsening public health (World Health Organization, 2024). Data-driven insights would further complement the design and optimization of bike-sharing schemes tailored to local contexts. This project is, therefore, meant to analyze the usage of the Capital Bikeshare system in Washington,

D.C. to provide actionable insights toward enhancing operational efficiency and user satisfaction. It intends to offer recommendations for implementing a successful bike-sharing program in Jordan to promote physical activity, improve urban mobility, and push towards environmental sustainability.

Project Scope

The scope of this project focuses on analyzing the usage patterns and trends of the Capital Bikeshare system from 2019 to 2024. The project will encompass data collection, processing, and analysis, with a particular focus on identifying key factors influencing bike rentals such as time of day, seasonality, temperature, and station popularity. The analysis will involve both descriptive and predictive methods, exploring temporal trends, user behavior, and the impact of weather on bike rentals. Key findings will be used to provide actionable recommendations for service improvement, including suggestions for enhancing accessibility, infrastructure, user experience, and sustainability. Additionally, the project will explore the feasibility of implementing a similar bike-sharing system in Jordan, proposing solutions tailored to local needs and challenges, including climate considerations, urban infrastructure, and the promotion of physical activity. The outcome of this project will be a comprehensive report that includes an analysis of current trends, recommendations for service optimization and system implementation in Jordan.

Importance of Data Analysis in Transportation Systems

In the present scenario of data, it is important to analyze transportation systems to comprehend the behavior and usage of these systems, as well as to make service delivery more efficient. The analysis of large datasets enables city planners and policymakers to make informed decisions regarding infrastructure development and resource allocation, as well as user

engagement strategies for the future. By analyzing the different aspects such as duration of rentals, peak usage times, and other external effects like the weather, trends can be derived to induce improvements in operations and user satisfaction.

Data analysis does not only result in an understanding of the current usage but also plays a significant part in predicting the future demands. Development and evolution of cities leave behind traces of past data and experiences, which can help inform the future use of transportation systems (Pucher & Buehler, 2017). This project focuses mainly on understanding usage patterns within Capital Bikeshare over a number of years so that the best solutions are derived for future applications, specifically in Jordan.

Objectives of the Study

The main purpose of this research would be to understand the patterns behind the use of the Capital Bikeshare system by analyzing rental usage data. This dataset includes rental durations, timestamps associated with stations from which a renter started and ended their rental, bike types, customer type (e.g., casual versus members), as well as some environmental data, including average temperature. Using these features, the present study aims to find out or reveal the operational dynamics of the bike-sharing system and how external influences affect the behavior of users. These specific objectives are given below:

1. **Analyzing Patterns of Use:** Analyze patterns of Capital Bikeshare being used concerning weather conditions and bike type to understand peak demand times, station popularity, and user preferences for different bike types.
2. **Evaluate Daily, Quarterly, and Annual Trends:** Evaluate daily, quarterly, and yearly trends in usage to identify peak usage times and stations that are most popular to better utilize resources.

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3. **Weather Analysis:** Determine the effect of temperature on bike-sharing riders as a way of knowing how weather influences user behavior. This can help for further decision-making for station management and resource distribution.
4. **Assess Bike Type Preference:** Review how different users consume different bike types, say the classic versus the e-bikes, giving insight into the users' requirements and improving the services.
5. **Time Series Analysis:** Time series analysis methods can be used to predict future demand for rentals and determine how usage patterns may change over time.
6. **Recommendations for Service Improvement:** Provide recommendations to enhance and facilitate implementation of bike-sharing systems in countries like Jordan by analyzing Capital Bikeshare usage patterns.

Overview of Capital Bikeshare

The key processes of the Capital Bikeshare system consist of various operational and service components, including:

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1. **Membership and Pricing:** Membership offers single rides, daily passes, and an annual Capital Bikeshare pass. Users pay a small fee to rent a bike for single rides, while all

other pricing schemes are geared toward loyalty to the program. For example, paying a flat fee for a day or an annual fee for unlimited short trips.

	Single ride	Day pass	Capital Bikeshare
	$0.05/min	$8/day	$95/year
	Get the app →	Get a day pass →	Join →
Bike unlocks	$1	Free	Free
Classic bike prices	$.05/min	45 mins free, then $.05/min	45 mins free, then $.05/min
Ebike prices	$0.15/min	$0.10/min	$0.10/min
Bike Angels			●

Figure 1. Capital Bikeshare Membership Pricing

2. **Bicycles and Locations:** The system comprises thousands of bicycles, with more than 600 docking stations across the city. Users only need to find and rent available bikes, either using a smartphone application or directly from the docking stations. The app also gives real-time information about availability of the bikes and station locations.
3. **Rent a Bicycle:** Rent a bike by doing either of the following:
    - Using the Capital Bikeshare app to find the nearest station with available bikes, then unlock one using any of the three membership passes.
    - Go to a docking station, select the desired bike and rent it using either a credit card or membership pass at the kiosk.
4. **Return Process:** After use, return the rented bike at a docking station that has open slots within the stipulated time limit; otherwise, extra fees will be incurred. The docking

stations are equipped with locking mechanisms that lock the bike once it has been returned.

5. **Integration with Public Transit:** The design of Capital Bikeshare was to ensure it links up with existing public transport systems by providing the appropriate infrastructure such that a person can easily ride while travelling. This will serve to reduce short-distance trips in cars, reducing road congestion and improving air quality.
6. **Data-Driven Insights:** The system collects data on usage behavior such as how long bikes are borrowed, what time of day use has the highest usage, and user demographic information. This information is extremely valuable to city planners and policymakers to enhance their efforts to gain an optimal distribution of bicycles and maximum efficiency.

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Literature Review

Bike-sharing has emerged as an effective solution in the development of public transport suggestive of a transition towards more sustainable travel options that cities have been able to consider to supplement cheap and environmentally sensitive means of moving people. Services like Capital Bikeshare in Washington, D.C., have been studied for their operational success, highlighting key factors that influence ridership and system effectiveness. The literature review examines some existing studies with the aim of providing an overview of essential elements surrounding bike-sharing systems as well as their interfaces with urban mobility and future convenience in replicating such a system in Jordan.

Temporal Trends in Bike Rentals

One of the primary focuses of this analysis is to explore temporal trends in bike rentals. Research shows that bike-sharing is mostly affected by seasons and weather (JDAntos, 2012). For example, higher temperatures are usually associated with greater use of the rented bicycles with more traveling by those preferring cycling use. On the other hand, lower temperatures and rainy days witness decreased usage.

Understanding such seasonal differences is important in managing resources. For example, if bike rentals were found to peak during certain months or certain days of the week, planners might ensure that they put in place sufficient resource capacity to match this demand. Also, analyzing behavior patterns among users can be instrumental in enhancing the optimization of station placements and improving service efficiency overall.

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Weather's Impact on Usage Patterns

The weather conditions are an important factor which dramatically affects the usage of bikeshare services. This study will investigate the effects of different weather variables on rental rates. For example, in periods of low temperatures and heavy rains, it is less likely that users will rent bikes due to discomfort and safety issues (JDAntos, 2012). By analyzing historical weather data and rental statistics, it would lead to the quantification of these effects and yield insights for service improvement during adverse weather conditions.

Such knowledge could be used for purposes of marketing strategies like tailoring campaigns motivating people to bike during favorable weather conditions or giving incentives during low traffic times, such as discounts.

User Behavior and Membership Types

Bike-sharing systems' usage can usually be identified by the difference between short-term users (e.g. tourist bikers) and long-term users (e.g. commuters). According to research, these groups are different in terms of usage pattern, with the ongoing users more likely to rent than casual users (Kaviti, Venigalla, & Lucas, 2019). Such aspects have strong implications on the system, which might require especially on a long-time basis differing type of services such as discounted membership or priority access of bikes at peak hours. Thus, it would be beneficial if both have a good understanding of users' needs for a new bike-share to be accessible and attractive to a larger pool of potential users.

Popularity of Specific Stations

It is one of the other objectives of this research to discover which are the stations that have higher rent than others. The knowledge of station popularity will be a factor in the decision

making of a station's location or resources' allocation. For instance, it would sometimes be advisable to increase bicycle stocks at a particular station or even expand the station network in underserved areas when most of the visited stations show high visitor numbers at certain times of the day, week, or year.

An inventory of station use from different time and condition scenarios will help planners to make resource allocation decisions to be as precise as possible. Ideally, such precision would improve the user experience, and coupled with proper estimation of the need for resources, it would improve operational efficiency (Vishkaei, Mahdavi, Mahdavi-Amiri, & Khorram, 2020).

Recommendations for Service Improvement

The interpretations from this analysis will also pinpoint trends that will probably include recommendations for ensuring efficiency in the Capital Bikeshare system. Understanding the behavior of a user in peak demand periods or on specific user-preferred routes allows planners to install strategies to make available bikes during high-demand periods.

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Recommendations could include targeted marketing campaigns to stimulate rentals during off-peak times or enhancing customer experiences during bad weather. For instance, making users more comfortable on rainy days will include providing covered waiting areas at popular stations.

Broader Implications for Urban Transportation

The research findings do not only extend to Washington, but also serve as an important reference for other cities that may be interested or are already operating bikeshare programs. A bike-sharing system for Jordanians has a great chance of transforming urban transport (Shbeed, 2018), as well as enhancing public health through physical activity (Bauman, Crane, Drayton, &

Titze, 2017). Cycling will automatically be part and parcel of daily life, which will seal several challenges facing the country, amongst them include road congestion, air pollution levels, and sedentary lifestyles.

Promoting Physical Activity

One of the prominent advantages of these bike-sharing programs is encouraging their users to engage in physical activity. It is well established that one can incorporate exercise into an individual's routine through daily cycling, especially within the urban setting, where sedentary lifestyles encourage it more (World Health Organization, 2024). Cyclists in Jordan, wherein the number of health issues related to lifestyle such as obesity and cardiovascular diseases seem to be on the rise (Ajlouni, Khader, Batieha, Jaddou, & El-Khateeb, 2020), can thus benefit substantial public health dividends by initiating cycling through a bike-sharing facility. Studies show that bike-sharing can significantly increase the amount of physical activity enjoyed by users (Bauman, Crane, Drayton, & Titze, 2017). For instance, in Colorado, the reduced amount of premature deaths and disability-adjusted life years was due to increasing physical activity levels among the population, amounting to millions of dollars saved in healthcare costs from adopting the programs (The 3 Health Impacts of a Bike Share System for Cities, 2021)

Bike-sharing makes it easier to gain access to bicycles, which would encourage more people to cycle rather than take motorized transport for short trips, thereby integrating exercise into their daily activities.

Environmental Benefits

Bicycles definitely lead to increase in physical activity in individuals, but in addition to that, they contribute to sustainable environmental health as well (Pucher & Buehler, 2017). They

can lessen dependence on cars for short trips, thereby reducing greenhouse gas emissions and improving air quality (Fishman, Washington, & Haworth, 2014).

In relation to rapid growth in cities with an increasing number of vehicles, the benefits of having a car are many in terms of the reduced vehicle emissions that are not only good for the environment but also aims for public health benefits associated with less air pollution, which remains to be linked to respiratory diseases and other health problems.

Enhancing Urban Mobility

A bike-sharing system can also contribute to improving urban mobility by implementing the means to provide flexible and efficient transport options within existing public transit networks. Those short travels can be complemented with bike-shares to alleviate public transport demand burden and provide freedom from the bottleneck-laden road infrastructure fills. This is quite significant considering the population in Jordan's growing cities, wherein traffic congestion becomes more dramatic each day. Bike-sharing systems can contribute to urban transport networks' efficiency by lowering average trip durations as well as bringing additional options of transport to users (Yang, Cheng, Chen, & Wang, 2017).

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For example, people accustomed to being in a car or a bus in the case of a short trip will tend to shift to bicycle use, which will be much better for the traffic conditions and public transport users will spend less time waiting.

Infrastructure Development

In terms of infrastructure, a bike-sharing program would require dedicated bike lanes, safe parking facilities, and docking stations that are intended to encourage people to use the service (Buehler & Buck, 2011). The environment must be bike friendly, which, will go hand in

hand with safety, the most important consideration in ensuring that it encourages more and more people to adopt this transport mode. Only with the bike-sharing system well-connected with other public transit services can it create seamless travel experiences for users. Such multimodal arrangements enable individuals to combine biking with other travel modalities. Urban travel is made easier without resorting to cars.

Considerations for Implementing a Bikeshare System in Jordan

Implementing a bike-sharing system in Jordan requires considering several local factors, including urban infrastructure, cultural attitudes toward cycling, and the country's climate. Studies on an adoption of bikeshare in new cities emphasize that the provision of bike infrastructure must be integrated into current public transport systems because only when cycling is perceived as safe and easy will people resort to it as a mode of transport (Buehler & Buck, 2011). In a country like Jordan wherein cycling has not been yet widely adopted by citizens, it is evident that public education and awareness campaigns will be instrumental in promoting the benefit and advantages of these bikeshare systems to its users as well as pushing non-adopters to habitual use of cycling as a daily activity.

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The local climate of Jordan has further posed some unique challenges and opportunities for implementation of bikeshare system. Summers are hot and winters are mild, which means the weather could affect the utilization and has similar effects to other cities when it comes to seasonality. Hence, it should be those areas where demand is constant along the demand curve throughout the year, for example, near university campuses, commercial areas, or tourist attractions. Besides, the durability of the bicycle and its suitability for the climatic condition should be taken into account for sustainability purposes.

Conclusion

All in all, bike-sharing scheme implementation in Jordan has manifold benefits from merely making it convenient to travel between points A to B (Capital Bikeshare, n.d.). It also promotes physical activity (Bauman, Crane, Drayton, & Titze, 2017), improves urban mobility (Fishman, Washington, & Haworth, 2014), and encourages environmental sustainability (Pucher & Buehler, 2017). This scheme is significant in addressing contemporary public health and transportation issues in Jordan, especially lifestyle-related health problems, which are rampant in urban environments. Research indicates that simply cycling is a very blatant way to exercise and can even be incorporated into daily routine activities resulting in improved health outcomes (Fraser & Lock, 2010). The most important aspect is to make access to bicycles easy and simple so that hopefully more people will get into the habit of choosing biking rather than motorized forms of transport for short trips. A bike-sharing program will contribute toward creating healthier lifestyles because it will certainly get more people on bikes for short trips rather than using motorized forms of transport. By understanding these dynamics, cities like Jordan can develop a bikeshare system tailored to local needs and conditions.

Notably, the analysis of the Capital Bikeshare system usage patterns in 2019 to 2024 observes how data-driven decision making finds its importance in urban public transportation systems. This study is aimed at actionable insights that can be used to enhance operational efficiency and user satisfaction characterized by an examination of temporal trends and external factors impacting bike rentals. These innovations will continue adding bicycle-sharing systems as modern cities evolve into healthier and green metropolis environments for all their citizens. Successful implementation of such systems is expected to reduce road congestion and improve air quality while further promoting alternative mobility as an option for increasingly more

commuters to have active transportation that is safer than just being a simple form of public commuting.

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Methodology

This project will employ a systematic approach towards understanding the patterns and factors influencing bike rentals. The data will be analyzed in different statistical methods to extract trends, relationships, and insights. The analysis would focus on how different factors, such as time and weather, affect bike usage. In addition, descriptive statistics, correlation analysis, and others will be applied to the dataset. Furthermore, visualizations will be also done to communicate the findings effectively and show important features and trends observed in the data.

Data Collection

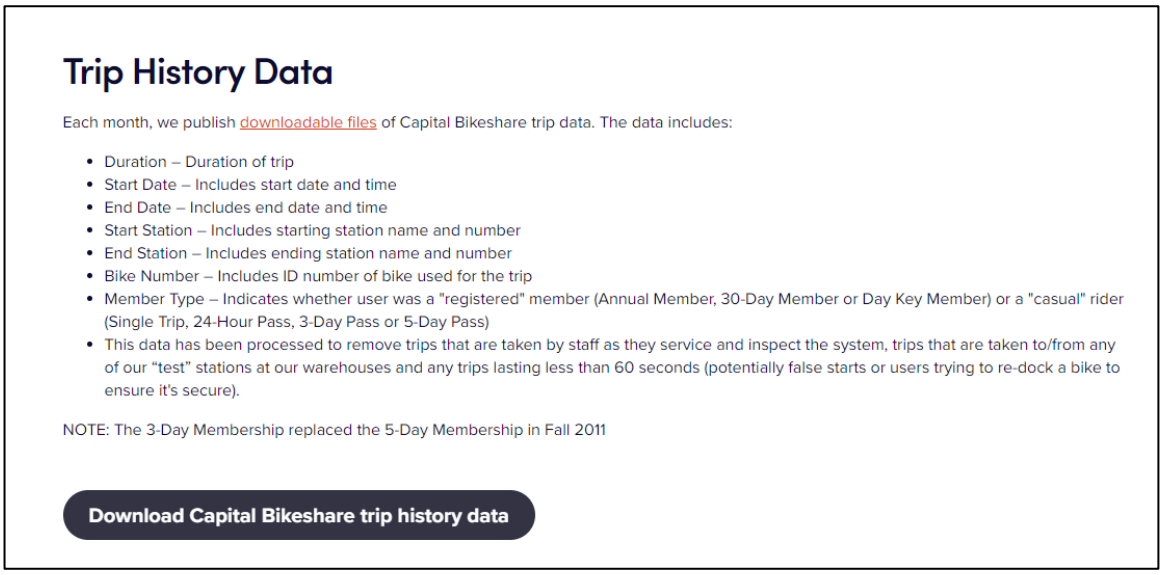
Data collection provides the foundation for any analysis, enabling informed decision-making and the development of reliable insights. Its purpose is to gather relevant and reliable information that will help analyze and predict the potential success of a bike-share system in Jordan. This project will primarily rely on two key sources for analysis.

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Data Sources

1. **Capital Bikeshare Dataset:** This data was downloaded from the official Capital Bikeshare website[1], which provides publicly available information on bike-share usage. The dataset includes variables such as station information, ride duration, start and end times, and bike types used.

[1] Capital Bikeshare System Data retrieved from https://capitalbikeshare.com/system-data



Trip History Data

Each month, we publish downloadable files of Capital Bikeshare trip data. The data includes:

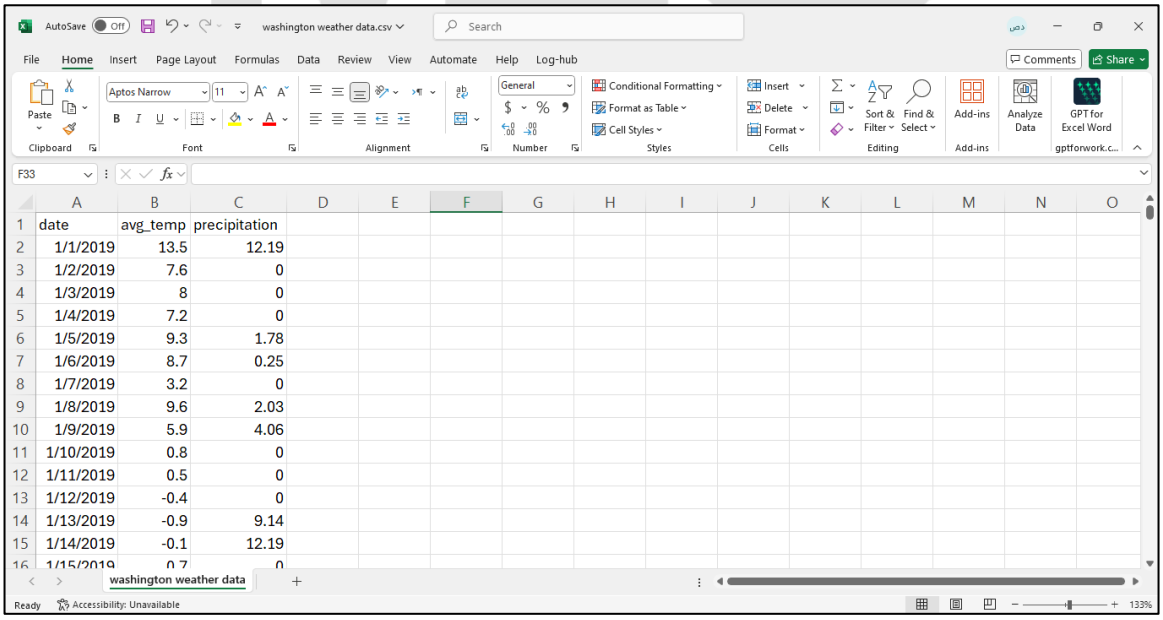
- Duration – Duration of trip
- Start Date – Includes start date and time
- End Date – Includes end date and time
- Start Station – Includes starting station name and number
- End Station – Includes ending station name and number
- Bike Number – Includes ID number of bike used for the trip
- Member Type – Indicates whether user was a "registered" member (Annual Member, 30-Day Member or Day Key Member) or a "casual" rider (Single Trip, 24-Hour Pass, 3-Day Pass or 5-Day Pass)
- This data has been processed to remove trips that are taken by staff as they service and inspect the system, trips that are taken to/from any of our "test" stations at our warehouses and any trips lasting less than 60 seconds (potentially false starts or users trying to re-dock a bike to ensure it's secure).

NOTE: The 3-Day Membership replaced the 5-Day Membership in Fall 2011

Download Capital Bikeshare trip history data

Figure 2. Capital Bikeshare System Data

2. **Weather Data:** Weather data for Washington, D.C. was collected by copying and pasting historical weather records, including daily average temperature and precipitation, from Weather Underground[2] directly into an Excel sheet.



	A	B	C
1	date	avg_temp	precipitation
2	1/1/2019	13.5	12.19
3	1/2/2019	7.6	0
4	1/3/2019	8	0
5	1/4/2019	7.2	0
6	1/5/2019	9.3	1.78
7	1/6/2019	8.7	0.25
8	1/7/2019	3.2	0
9	1/8/2019	9.6	2.03
10	1/9/2019	5.9	4.06
11	1/10/2019	0.8	0
12	1/11/2019	0.5	0
13	1/12/2019	-0.4	0
14	1/13/2019	-0.9	9.14
15	1/14/2019	-0.1	12.19
16	1/15/2019	0.7	0

Figure 3. Washington, D.C. Weather Data

[22] Weather Data retrieved from https://www.wunderground.com/weather/us/dc/washington

Data Cleaning and Preprocessing

The data will be preprocessed for accuracy and consistency. Discarding the irrelevant columns and standardizing the date columns, the data will be more effective than before in analysis. Duplicate entries will be removed, while missing value entries will be imputed with proper statistical methods. Merging of appropriate weather details according to date should enhance the data analysis. Outliers will be statistically identified and replaced by a median value to take care of the integrity of the data. Finally, the cleaned data will be saved, ready for exploration and visualization.

Comprehensive Data Analysis

This analysis will explore the usage patterns of the Capital Bikeshare system, focusing on various factors that influence bike rentals. This project intends to analyze across different time frames: rentals over the year, over a week, and changes in hourly rentals to discover how time-related patterns differ. It will also show how weather conditions, for instance, temperature, interact with the bike usage. The popularity of bike stations and the two market segments for the different bike types will also be investigated. All these analyses hope to bring an understanding of the dynamics behind bike rental.

Key Research Questions

To guide this analysis, several key research questions will be addressed:

1. How do seasonal variations affect bike rental patterns?
2. What is the impact of weather conditions on bike rentals?
3. Which stations are most popular among users?
4. How can insights from this analysis inform recommendations for service improvement, and implementing a bike-sharing system in Jordan?

Time-Series Forecasting

Time-series forecasting involves the use of historical data to make informed predictions concerning future occurrences or trends. In the context of this analysis, time-series forecasting will seek to identify bike rentals over time. With the past rental data in hand, this project intends to develop a predictive model that will estimate future demands for bicycles based on past rental patterns. These insights will help make more accurate forecasts, helping to anticipate future demand, optimize bike availability, and help decision-makers better manage resources.

Interactive Dashboard

Interactive dashboards created with Tableau will provide stakeholders with an intuitive and user-friendly platform from which to view new trends in bike rentals, weather patterns, and station usage. Such dashboards will allow users to filter and drill down data to very specific time periods, weather conditions, or even stations, allowing them to closer look at the associated patterns and factors that influence bike rental usage. Stakeholders, such as city planners or transportation authorities, would be able to make informed decisions based on real-time visualizations to better understand bike-sharing trends and optimize the system for service improvement and resource allocation.

Recommendations for Implementing a Bike-Sharing System in Jordan

The findings from these analyzes will feature recommendations geared towards optimizing a bike-sharing system in Jordan. This entails discussing factors such as station placement, usage according to weather conditions, and how to improve overall system efficiency. All of these recommendations will be directed toward enhancing the experience from the user's end, maximizing usage of the bike-sharing system, and making the system more convenient and sustainable.

Technical Requirements

These are the necessary technical requirements necessary for a successful execution of the project, accurate analysis and visualizations of trends, and building predictive models to optimize the bike-sharing system in Jordan:

1. **Data Collection and Storage:** Access to the Capital Bikeshare dataset, and weather data in a structured format, such as CSV, or Excel.
2. **Data Processing:** Use of R for data analysis, including data cleaning and preprocessing, integrating weather data with bike usage data to identify correlation, and transformation of time-series data for forecasting. Libraries include, but are not limited to:
    - forecast for time-series forecasting
    - dplyr for data manipulation
    - ggplot2 for data visualization
    - lubridate for handling time-related data
3. **Predictive Models:** Forecast future bike rental demand using historical data. This is achieved through time series analysis, which is a statistical method used to analyze data points collected over time to identify patterns, trends and seasonality.
4. **Dashboard and Visualization:** Development of interactive dashboards in Tableau to visualize, filter, and explore data.
5. **Documentation:** Documentation of the methodology, code, and analysis results, and preparation of actionable recommendations for implementing a bike-sharing system in Jordan based on findings.

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Data Analysis

Data Cleaning & Preprocessing

Data cleaning is a very important step in training a dataset prior to performing any analyzes on that data. Cleaning the data includes determining the missing values, inconsistencies, and errors, ensuring the data is accurate, complete, and reliable. Clean data gets more useful insights and better decisions, so it is an essential step into any data analysis. Here are the steps involved in the data cleaning process:

Combining the Data

1. **Importing the Data:** The Capital Bikeshare datasets were unzipped and placed in a single directory, then loaded into R for analysis. This step is essential to ensure the data is organized and available for processing.

```
1. #insert path to the directory which includes all 70 files
2. file_list <- list.files(path = "C:\\Users\\PC\\Desktop\\\Capital Bikeshare Datasets",
pattern = "*.csv", full.names = TRUE)
```

Code 1. Importing the Data

2. **Mapping the Data:** The 70 separate files from January 2019 to October 2024 were merged into a single dataset. During this process, it was essential to map the changes in the dataset format that occurred in April 2020 by creating associations between two sets of items, such that each element in one set corresponds to an element in another set. The format shift required renaming columns to match the new structure, standardizing date and time formats, and ensuring that variables from both the pre- and post-April 2020 datasets were aligned.

```r
#map old column names to new column names
column_mapping <- c(
"start_date" = "started_at",
"end_date" = "ended_at",
"start_station_number" = "start_station_id",
"start_station" = "start_station_name",
"end_station_number" = "end_station_id",
"end_station" = "end_station_name",
"member_type" = "member_casual"
)
```

Code 2. Mapping the Data

3. **Initial Cleaning:** This code reads multiple files into dataframes, cleans and standardizes column names, and renames columns based on the mapping. It returns a list of cleaned dataframes.

```r
#all libraries have been installed using install.packages() prior to loading

library(readr)
library(janitor)

all_data <- lapply(file_list, function(file) {
  df <- read_csv(file)  #read each file individually into a dataframe with readr
  df <- clean_names(df)  #standardize column names by converting them to lowercase and removing spaces with janitor
  df <- df %>%
    rename_with(~ column_mapping[.x], .cols = intersect(names(df), names(column_mapping))) #rename columns based on the mapping done in the previous step
  return(df)  #return cleaned dataframe
```

```
12. })
```

Code 3. Initial Cleaning

4. **Combining Dataframes:** The dataframes have been efficiently combined into one.

```
1. library(dplyr)
2.
3. #combine all dataframes into one with dplyr
4. combined_df <- bind_rows(all_data)
```

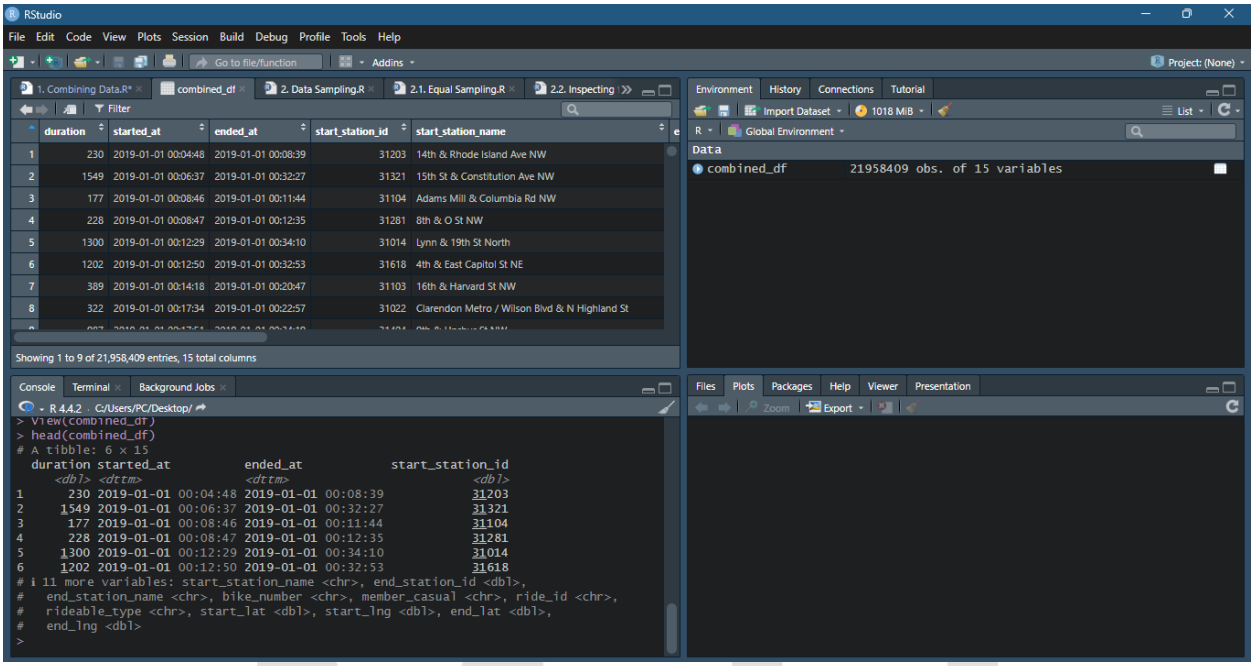
Code 4. Combining Dataframes

The total number of observations was over 22 million before sampling

5. **Saving & Loading the .csv File:** The combined dataframe has been written to a .csv file for future use and can later be retrieved by reading the same file back into R.

```
1. library(readr)
2.
3. #save & load with readr
4. #same method will be applied in future code to save and retrieve data from .csv files as
needed
5. write_csv(combined_df, "C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\1.
Combined Data\\combined_df.csv")
6. combined_df <- read_csv("C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\1.
Combined Data\\combined_df.csv")
```

Code 5. Saving & Loading



Snapshot 1. Combining the Data

Data Sampling

1. **Random Sample (General Sampling):** Random sampling was used to enable an analysis of general trends without focusing on month-to-month consistency. A simple random sample will select 1% of the rows from the entire dataset, regardless of the month.

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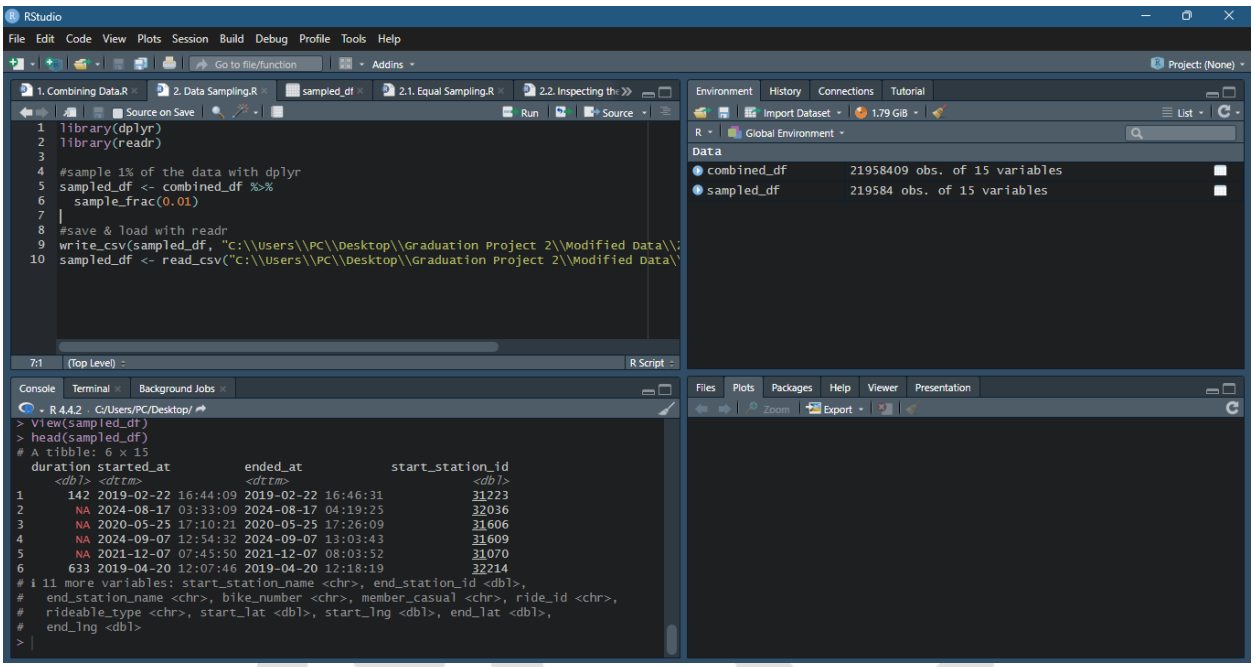
```
library(dplyr)
library(readr)

#sample 1% of the data with dplyr
sampled_df <- combined_df %>%
  sample_frac(0.01)

#save & load with readr
write_csv(sampled_df, "C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\2. Sampled Data\\sampled_df.csv")
```

```
10. sampled_df <- read_csv("C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\2.
Sampled Data\\sampled_df.csv")
```

Code 6. General Sampling



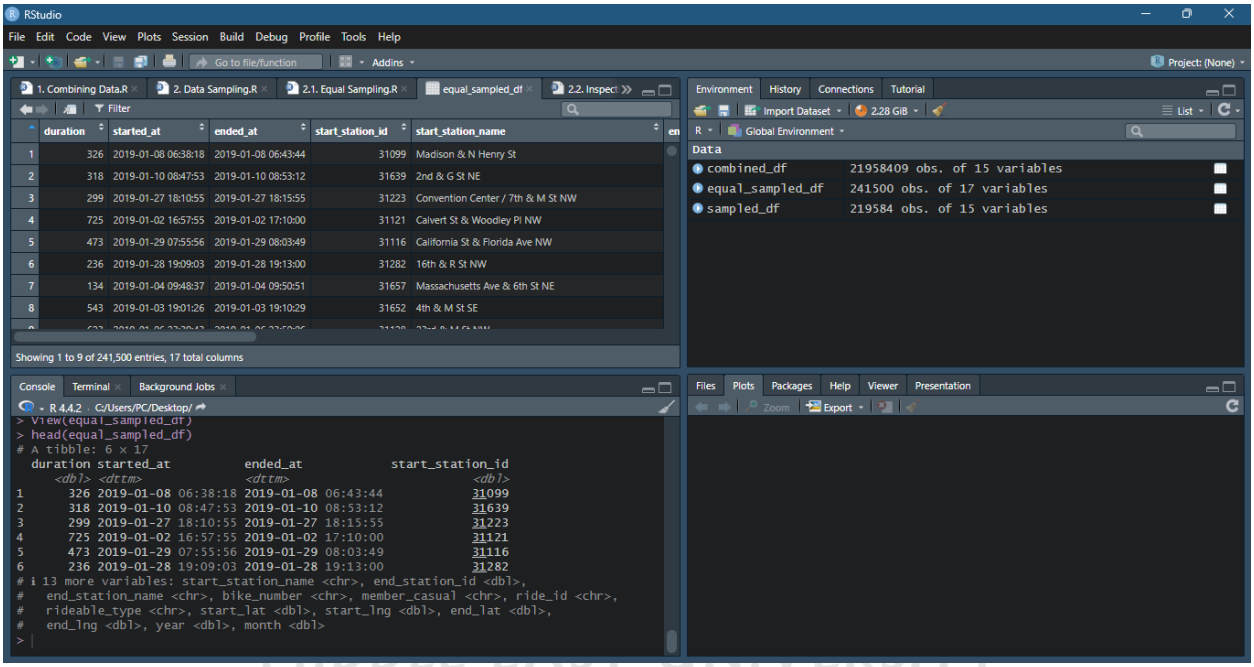
Snapshot 2. General Sample

2. **Sample of Equal Observations from Each Month (Stratified Sampling):** Stratified sampling was used to ensure balanced representation from each month, allowing for an analysis of monthly trends, patterns, and comparisons.

```
1. library(dplyr)
2. library(lubridate)
3. library(readr)
4.
5. equal_sampled_df <- combined_df %>%
6.   group_by(year = year(started_at), month = month(started_at)) %>% #group the dataset by
the year and month of the started_at column with lubridate
7.   sample_n(3500, replace = TRUE) %>% #sample 3,500 rows for each month of each year with
dplyr
8.   ungroup() #returning the dataset to its original ungrouped state
```

```
 9.
10. #save & load with readr
11. write_csv(equal_sampled_df, "C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified
Data\\2. Sampled Data\\equal_sampled_df.csv")
12. equal_sampled_df <- read_csv("C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified
Data\\2. Sampled Data\\equal_sampled_df.csv")
```

Code 7: Stratified Sampling



Snapshot 3. Stratified Sampling

Inspecting the Data

The structure of the dataset has been examined using head(), summary(), and str() functions. These functions provide an overview of the data, allowing us to identify any inconsistencies or potential issues early on.

```
1. library(naniar)
2.
3. str(sampled_df)  #check the structure and types of columns
4. summary(sampled_df)  #get a statistical summary of numerical columns
5. head(sampled_df)  #view the first few rows
```

```
6. gg_miss_var(sampled_df) #visualize the missing data across the columns in a graphical
format with naniar
```

Code 8: Data Inspection using head(), summary(), and str()

1. **Examining the Data using head(), summary(), and str():** These functions provide an overview of the data, allowing us to identify any inconsistencies or potential issues early on. The following output was generated in console after running the previous code:

```
 1. > str(sampled_df)  #check the structure and types of columns
 2. spc_tbl_ [219,584 × 15] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 3.  $ duration          : num [1:219584] 142 NA NA NA NA 633 NA NA NA NA ...
 4.  $ started_at        : POSIXct[1:219584], format: "2019-02-22 16:44:09" "2024-08-17
03:33:09" "2020-05-25 17:10:21" ...
 5.  $ ended_at          : POSIXct[1:219584], format: "2019-02-22 16:46:31" "2024-08-17
04:19:25" "2020-05-25 17:26:09" ...
 6.  $ start_station_id  : num [1:219584] 31223 32036 31606 31609 31070 ...
 7.  $ start_station_name: chr [1:219584] "Convention Center / 7th & M St NW" "Fenton St &
Ellsworth Dr" "Potomac & Pennsylvania Ave SE" "Maine Ave & 7th St SW" ...
 8.  $ end_station_id    : num [1:219584] 31281 NA 31606 31628 31293 ...
 9.  $ end_station_name  : chr [1:219584] "8th & O St NW" NA "Potomac & Pennsylvania Ave SE"
"1st & I St SE" ...
10.  $ bike_number       : chr [1:219584] "W01436" NA NA NA ...
11.  $ member_casual     : chr [1:219584] "Member" "casual" "member" "casual" ...
12.  $ ride_id           : chr [1:219584] NA "3E3709917995B650" "27B3019E5B57D24E"
"6F70967C500EE90A" ...
13.  $ rideable_type     : chr [1:219584] NA "electric_bike" "docked_bike" "classic_bike" ...
14.  $ start_lat         : num [1:219584] NA 39 38.9 38.9 38.9 ...
15.  $ start_lng         : num [1:219584] NA -77 -77 -77 -77.1 ...
16.  $ end_lat           : num [1:219584] NA 39 38.9 38.9 38.9 ...
17.  $ end_lng           : num [1:219584] NA -77 -77 -77 -77.1 ...
18.  - attr(*, "spec")=
```

```
  .. cols(
  ..   duration = col_double(),
  ..   started_at = col_datetime(format = ""),
  ..   ended_at = col_datetime(format = ""),
  ..   start_station_id = col_double(),
  ..   start_station_name = col_character(),
  ..   end_station_id = col_double(),
  ..   end_station_name = col_character(),
  ..   bike_number = col_character(),
  ..   member_casual = col_character(),
  ..   ride_id = col_character(),
  ..   rideable_type = col_character(),
  ..   start_lat = col_double(),
  ..   start_lng = col_double(),
  ..   end_lat = col_double(),
  ..   end_lng = col_double()
  .. )
 - attr(*, "problems")=<externalptr>
```

Figure 4. Using str() to Inspect the Data

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The general sample contains 219,584 rows and 15 columns, with various data types including numeric, character, and datetime. Some columns, like duration, start_lat, and end_lat, have missing values (NA), indicating incomplete data. The dataset includes temporal, geographic, and categorical information for bike rides, such as ride duration, start and end locations, and bike types.

```
> summary(sampled_df)  #get a statistical summary of numerical columns
    duration        started_at                      ended_at
Min.   :   60    Min.   :2019-01-01 00:34:56.00   Min.   :2019-01-01 00:38:11.00
1st Qu.:  389    1st Qu.:2020-12-20 14:46:01.25   1st Qu.:2020-12-20 15:09:08.50
Median :  670    Median :2022-09-28 15:10:20.00   Median :2022-09-28 15:35:09.50
Mean   : 1035    Mean   :2022-06-16 08:59:21.13   Mean   :2022-06-16 09:19:57.97
3rd Qu.: 1136    3rd Qu.:2024-01-30 21:17:08.75   3rd Qu.:2024-01-30 21:36:43.25
```

```
 Max.   :86289    Max.   :2024-10-31 23:46:24.00   Max.   :2024-10-31 23:58:33.00
 NA's   :180240
 start_station_id start_station_name end_station_id  end_station_name   bike_number
 Min.   :    0    Length:219584      Min.   :    0   Length:219584      Length:219584
 1st Qu.:31208    Class :character   1st Qu.:31213   Class :character   Class :character
 Median :31276    Mode  :character   Median :31275   Mode  :character   Mode  :character
 Mean   :31364                       Mean   :31366
 3rd Qu.:31604                       3rd Qu.:31606
 Max.   :33200                       Max.   :33200
 NA's   :18719                       NA's   :19962
 member_casual        ride_id          rideable_type        start_lat       start_lng
 Length:219584      Length:219584      Length:219584      Min.   :38.76   Min.   :-77.37
 Class :character   Class :character   Class :character   1st Qu.:38.89   1st Qu.:-77.04
 Mode  :character   Mode  :character   Mode  :character   Median :38.90   Median :-77.03
                                                          Mean   :38.90   Mean   :-77.03
                                                          3rd Qu.:38.92   3rd Qu.:-77.01
                                                          Max.   :39.13   Max.   :-76.83
                                                          NA's   :39344   NA's   :39344
    end_lat         end_lng
 Min.   :11.80   Min.   :-77.49
 1st Qu.:38.89   1st Qu.:-77.04
 Median :38.90   Median :-77.03
 Mean   :38.90   Mean   :-77.03
 3rd Qu.:38.91   3rd Qu.:-77.01
 Max.   :39.13   Max.   :-55.85
 NA's   :39609   NA's   :39609
> head(sampled_df)  #view the first few rows
# A tibble: 6 × 15
  duration started_at          ended_at            start_station_id start_station_name
     <dbl> <dttm>              <dttm>                         <dbl> <chr>
```

```
38. 1      142 2019-02-22 16:44:09 2019-02-22 16:46:31           31223 Convention Center /
7th & M St …
39. 2       NA 2024-08-17 03:33:09 2024-08-17 04:19:25           32036 Fenton St & Ellsworth
Dr
40. 3       NA 2020-05-25 17:10:21 2020-05-25 17:26:09           31606 Potomac & Pennsylvania
Ave SE
41. 4       NA 2024-09-07 12:54:32 2024-09-07 13:03:43           31609 Maine Ave & 7th St SW
42. 5       NA 2021-12-07 07:45:50 2021-12-07 08:03:52           31070 Arlington Blvd &
Fillmore St
43. 6      633 2019-04-20 12:07:46 2019-04-20 12:18:19           32214 Reston Town Center
Transit Stat…
44. # ℹ 10 more variables: end_station_id <dbl>, end_station_name <chr>, bike_number <chr>,
45. #   member_casual <chr>, ride_id <chr>, rideable_type <chr>, start_lat <dbl>, start_lng
<dbl>,
46. #   end_lat <dbl>, end_lng <dbl>
```

Figure 5. Using summary() to Inspect the Data

This function displays the first six rows of the dataset, showing its structure and contents. It provides an overview of the variables and their data types for a quick inspection.

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Descriptive Statistics

From the above summary we can extract these observations:

- Median: 670 seconds (~11 minutes), Mean: 1035 seconds (~17 minutes).
- Max duration (86,289 seconds) suggests possible outliers.
- Significant NA values in duration, station IDs, start_lat/lng, and end_lat/lng.
- Latitudes and longitudes appear consistent but include missing values.

2. **Visualizing Missing Values:** The function gg_miss_var() visualizes the missing data in the columns. It creates a plot that shows which variables have missing values, helping to easily identify patterns of missingness across the dataset.

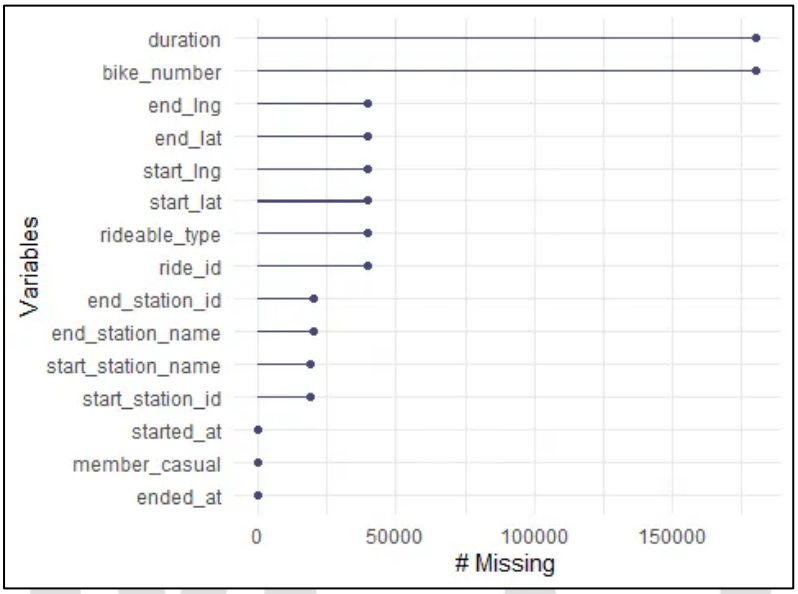


Figure 6. Visualizing Missing Values

The plot shows that there are varying degrees of missing data across the different variables.

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Data Cleaning

All the steps below were applied to both sample files.

1. **Removing Unnecessary Columns:** Columns not relevant to the analysis have been removed to simplify the dataset. This leads to improved processing efficiency and reduced computational load, making data handling and analysis more efficient.

```
1. library(dplyr)
2.
3. #remove unnecessary columns with dplyr
4. cleaned_df <- sampled_df %>%
5. select(-ride_id, -bike_number, -start_station_id, -end_station_id)
```

Code 9. Removing Unnecessary Columns

Removing ID columns during data cleaning is preferred because they typically do not add value to the analysis. They are unique identifiers that help distinguish individual records but do not provide predictive power or contribute to understanding relationships in the data.

2. **Verifying Proper Formatting of Date Columns as POSIXct**: Converted date-time strings into POSIXct objects, R's standard for handling date-time data. This standardization ensures the data is stored in a consistent format, making it easier to handle in further analysis, such as calculating time differences (e.g., trip duration) and performing other date-time operations.

```
#convert datetime columns into POSIXct
cleaned_df$started_at <- as.POSIXct(cleaned_df$started_at, format = "%Y-%m-%d %H:%M:%S")
cleaned_df$ended_at <- as.POSIXct(cleaned_df$ended_at, format = "%Y-%m-%d %H:%M:%S")
```

Code 10. Converting date-time data into POSIXct

3. **Checking for Duplicate Rows**: Duplicate observations have been removed to avoid skewed results, improve model accuracy, reduce redundancy, and ensure consistent and reliable insights.

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```
library(dplyr)

#check for duplicate rows with dplyr
cleaned_df <- cleaned_df %>%
distinct()
```

Code 11. Checking for Duplicate Rows

4. **Removing Observations with Missing Values:** Missing values in start_station_name and end_station_name columns have been removed because it's more efficient than imputing data for these categorical columns, unlike the numerical columns where imputation may be more feasible.

```
library(dplyr)
```

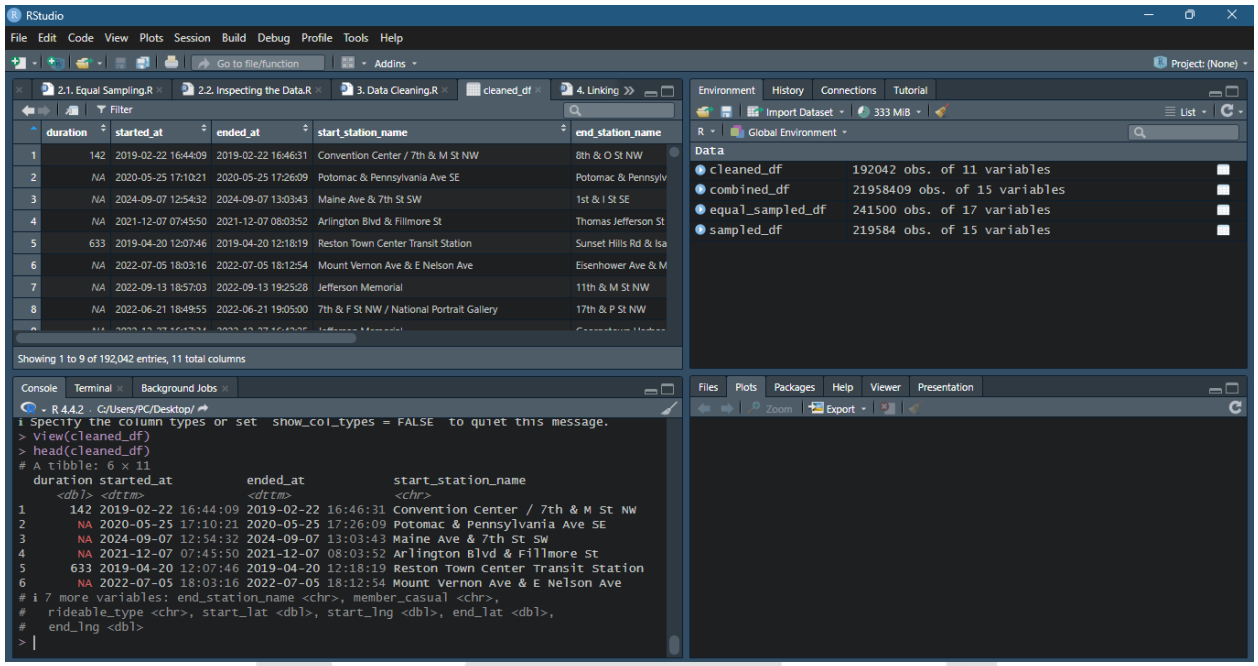
```
2.
3. #remove missing values in start_station_name and end_station_name with dplyr
4. cleaned_df <- cleaned_df %>%
5.   filter(!is.na(start_station_name), !is.na(end_station_name))
```

Code 12. Removing Observations with Missing Values

5. **Standardizing Categorical Data by Converting to Lowercase:** The member_casual column has been converted to lowercase, ensuring uniformity in the categorical data.

```
1. library(readr)
2.
3. #convert 'member_casual' column to lowercase
4. cleaned_df$member_casual <- tolower(cleaned_df$member_casual)
5.
6. #save & load with readr
7. write_csv(cleaned_df, "C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\3.
Cleaned Data\\cleaned_df.csv")
8. cleaned_df <- read_csv("C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\3.
Cleaned Data\\cleaned_df.csv")
9. equal_cleaned_df <- read_csv("C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified
Data\\3. Cleaned Data\\equal_cleaned_df.csv")
```

Code 13. Standardizing the Data



Snapshot 4. Data Cleaning

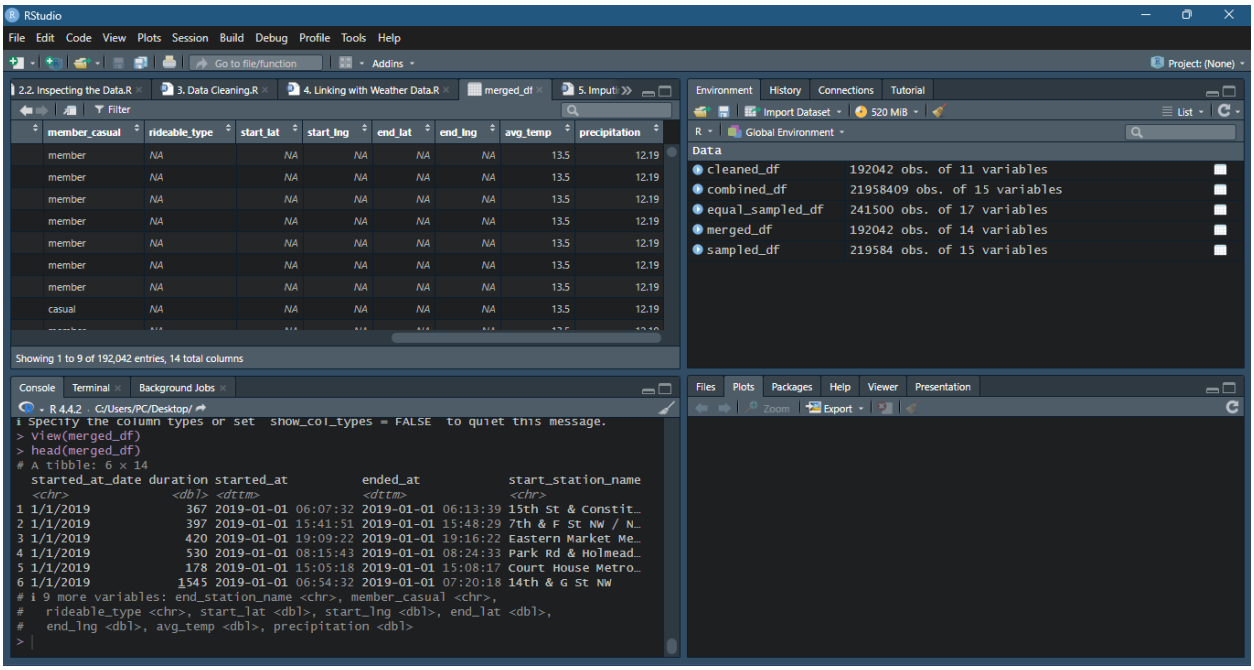
Linking with Weather Data

Data have been merged with web-scraped weather data (washington_weather_data) based on the started_at_date column. The merged data is saved to a .csv file for further use, along with an equally merged dataset (equal_merged_df) for comparison.

```
1. library(readr)
2.
3. #convert the date column in washington_weather_data into a date object in R
4. washington_weather_data$date <- as.Date(washington_weather_data$date)
5.
6. #create a new column 'started_at_date' that only contains the date part of 'started_at'
7. cleaned_df$started_at_date <- as.Date(cleaned_df$started_at)
8.
9. #merge the datasets by the new 'started_at_date' column
10. merged_df <- merge(cleaned_df, washington_weather_data, by.x = "started_at_date", by.y =
"date", all.x = TRUE)
11.
```

```
#save & load with readr
write_csv(merged_df, "C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\4. Merged Data\\merged_df.csv")
merged_df <- read_csv("C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\4. Merged Data\\merged_df.csv")
equal_merged_df <- read_csv("C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\4. Merged Data\\equal_merged_df.csv")
```

Code 14. Linking with Weather Data



Snapshot 5. Merging the Data

Imputing Missing Values

1. **Imputing Missing Values using Mean:** For numeric columns like start_lng, start_lat, end_lng, and end_lat, missing values have been replaced with the column's mean value.

```
library(dplyr)

#replace missing values with mean for numeric columns with dplyr
imputed_df <- merged_df %>%
  mutate(
```

```
    start_lng = ifelse(is.na(start_lng), mean(start_lng, na.rm = TRUE), start_lng),
    start_lat = ifelse(is.na(start_lat), mean(start_lat, na.rm = TRUE), start_lat),
    end_lng = ifelse(is.na(end_lng), mean(end_lng, na.rm = TRUE), end_lng),
    end_lat = ifelse(is.na(end_lat), mean(end_lat, na.rm = TRUE), end_lat)
  )
```

Code 15. Imputing Data using Mean

2. **Imputing Missing Values using Mode:** For the rideable_type column, missing values have been replaced with the mode.

```
#find the mode in the 'rideable_type' column
mode_value <- names(sort(table(imputed_df$rideable_type), decreasing = TRUE))[1]

#replace NA values in the 'rideable_type' column with the mode value
imputed_df$rideable_type[is.na(imputed_df$rideable_type)] <- mode_value
```

Code 16. Imputing Data using Mode

3. **Imputing Missing Values using Arithmetic Operations:** Missing values in the duration column have been calculated based on the time difference between ended_at and started_at.

```
library(readr)

#remove rows where the end time is earlier than the start time
imputed_df <- imputed_df %>%
  filter(ended_at >= started_at)

#calculate the duration in seconds (subtract end time from start time)
imputed_df$duration <- as.numeric(difftime(imputed_df$ended_at, imputed_df$started_at,
units = "secs"))

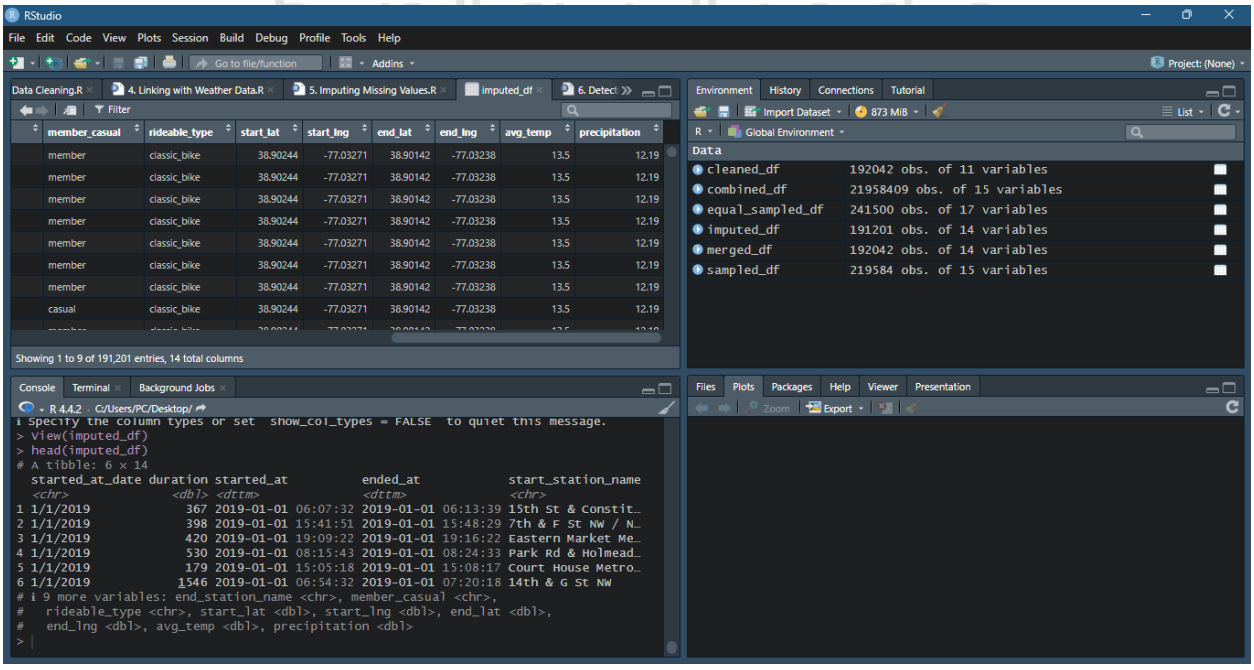
#impute missing durations by recalculating them for rows with NA
```

```
imputed_df$duration[is.na(imputed_df$duration)] <-
as.numeric(difftime(imputed_df$ended_at[is.na(imputed_df$duration)],
imputed_df$started_at[is.na(imputed_df$duration)], units = "secs"))

#remove rows where the duration is less than 20 seconds
imputed_df <- imputed_df %>%
  filter(duration >= 20)

#save & load with readr
write_csv(imputed_df, "C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\5. Imputed Data\\imputed_df.csv")
imputed_df <- read_csv("C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\5. Imputed Data\\imputed_df.csv")
equal_imputed_df <- read_csv("C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\5. Imputed Data\\equal_imputed_df.csv")
```

Code 17. Imputing Data using Arithmetic Operations



Snapshot 6. Imputing the Data

Detecting Outliers & Finalizing the Data

1. **Using Boxplots to Visualize Outliers:** Boxplots have been used to display the distribution of data, highlighting the median, quartiles, and potential outliers. The whiskers show the range of values, and points outside the whiskers are considered outliers. They provide a quick way to assess the spread and outliers in the data.

```
#boxplot for temperature column to visualize distributions
boxplot(imputed_df$avg_temp, main = "Temperature Boxplot")

#boxplot for precipitation column to visualize distributions
boxplot(imputed_df$precipitation, main = "Precipitation Boxplot")
```

Code 18. Visualizing Distributions

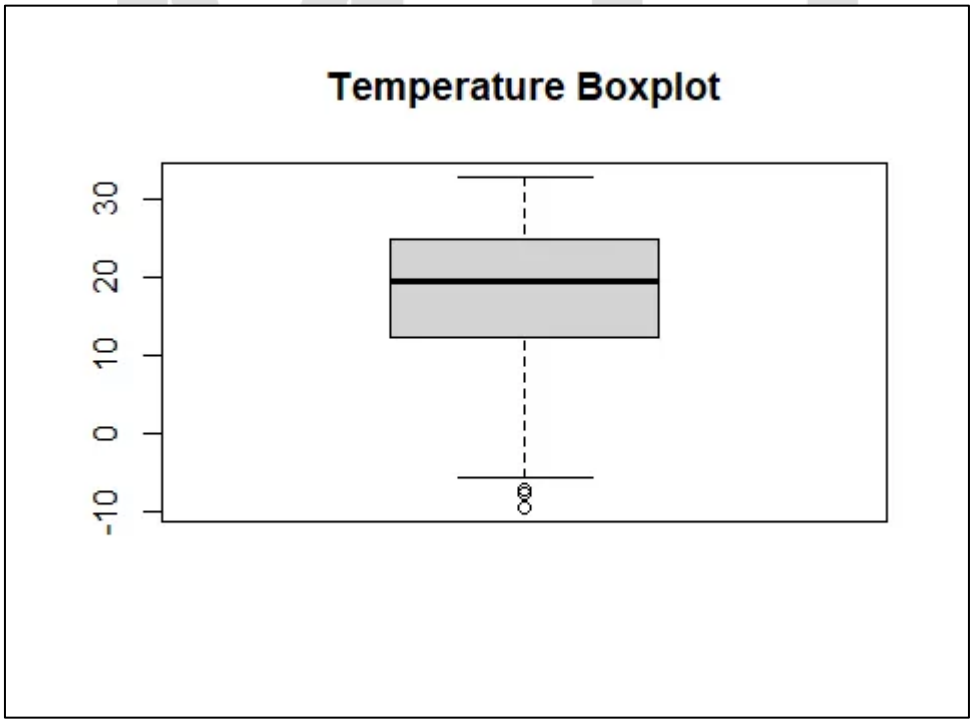


Figure 7. Temperature Boxplot

The boxplot reveals an outlier in the temperature data, located just below -10°C, as indicated by a point outside the whiskers.

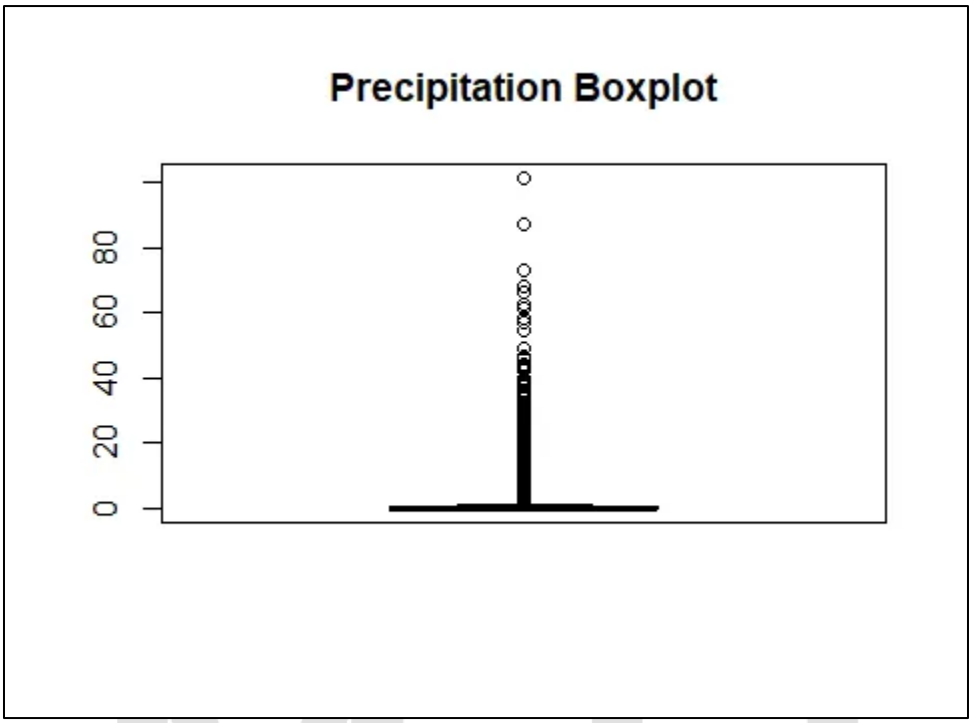


Figure 8. Precipitation Boxplot

The boxplot for precipitation looks unusual because the precipitation data is highly skewed, with most values clustered near zero, resulting in a flat IQR, and a few significant outliers, indicating unusually high precipitation values that fall outside the expected range.

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2. **Creating the IQR Outlier Detection Function:** The function calculates the first (Q1) and third (Q3) quartiles of the data, computes the IQR, and then identifies values outside the range defined by 1.5 times the IQR below Q1 and above Q3 as outliers.

```
#create a copy of the original dataset
final_df <- imputed_df

#function to detect outliers using IQR
detect_outliers_iqr <- function(x) {
  Q1 <- quantile(x, 0.25, na.rm = TRUE)
  Q3 <- quantile(x, 0.75, na.rm = TRUE)
  IQR <- Q3 - Q1
  lower_bound <- Q1 - 1.5 * IQR
```

```
  upper_bound <- Q3 + 1.5 * IQR
  return(x < lower_bound | x > upper_bound)
}
```

Code 19. Creating the IQR Outlier Detection Function

3. **Applying the IQR Function to the Data:** *The detect_outliers_iqr function has been applied to multiple numerical columns to detect outliers.*

```
#apply function to multiple numerical columns
outliers_avg_temp <- detect_outliers_iqr(final_df$avg_temp)
outliers_precipitation <- detect_outliers_iqr(final_df$precipitation)
outliers_start_lat <- detect_outliers_iqr(final_df$start_lat)
outliers_end_lat <- detect_outliers_iqr(final_df$end_lat)
outliers_start_lng <- detect_outliers_iqr(final_df$start_lng)
outliers_end_lng <- detect_outliers_iqr(final_df$end_lng)
outliers_duration <- detect_outliers_iqr(final_df$duration)
```

Code 20. Applying the Function

4. **Detecting Outliers using the IQR Function:** A logical vector has been created, which flags rows as outliers if any value in the specified columns is an outlier. If any of the columns contains an outlier, the corresponding row is marked as TRUE.

```
#combine outlier detection into one logical vector for all specified columns
outlier_rows_iqr <- outliers_avg_temp | outliers_precipitation | outliers_start_lat |
outliers_end_lat | outliers_start_lng | outliers_end_lng | outliers_duration
```

Code 21. Detecting Outliers

5. **Handling the Outliers:** Outlier values in the specified columns have been replaced with the median of the respective column.

```
library(readr)

#replace outliers with median
final_df$avg_temp[outliers_avg_temp] <- median(final_df$avg_temp, na.rm = TRUE)
```

```
final_df$precipitation[outliers_precipitation] <- median(final_df$precipitation, na.rm = TRUE)
final_df$start_lat[outliers_start_lat] <- median(final_df$start_lat, na.rm = TRUE)
final_df$end_lat[outliers_end_lat] <- median(final_df$end_lat, na.rm = TRUE)
final_df$start_lng[outliers_start_lng] <- median(final_df$start_lng, na.rm = TRUE)
final_df$end_lng[outliers_end_lng] <- median(final_df$end_lng, na.rm = TRUE)
final_df$duration[outliers_duration] <- median(final_df$duration, na.rm = TRUE)

#save & load with readr
write_csv(final_df, "C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\6. Final Data\\final_df.csv")
final_df <- read_csv("C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\6. Final Data\\final_df.csv")
equal_final_df <- read_csv("C:\\Users\\PC\\Desktop\\Graduation Project 2\\Modified Data\\6. Final Data\\equal_final_df.csv")
```

Code 22. Handling the Outliers

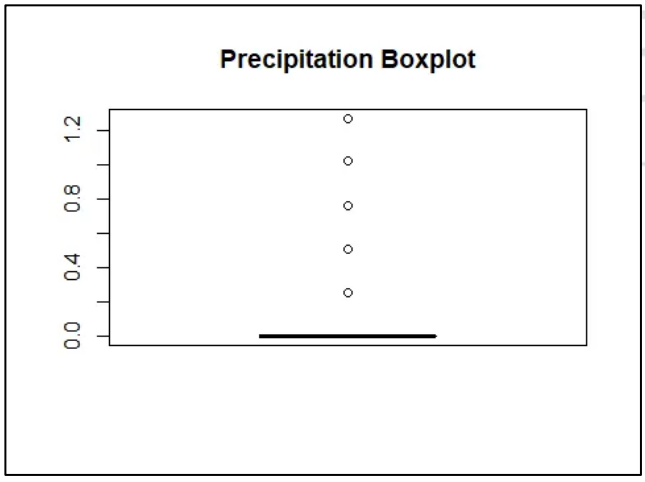


Figure 9. Precipitation Boxplot After Removing Outliers

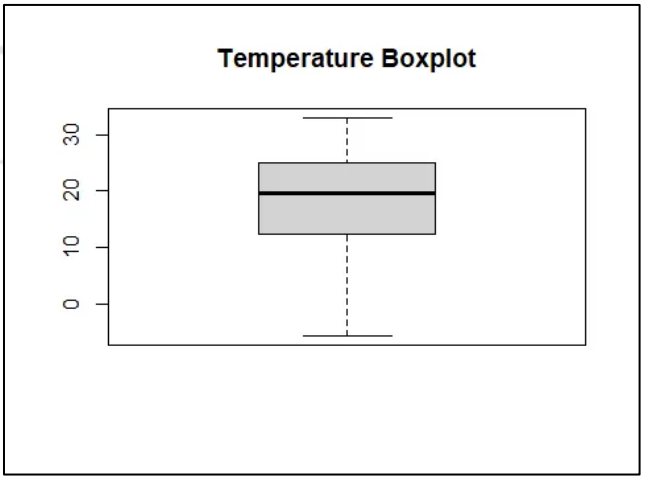
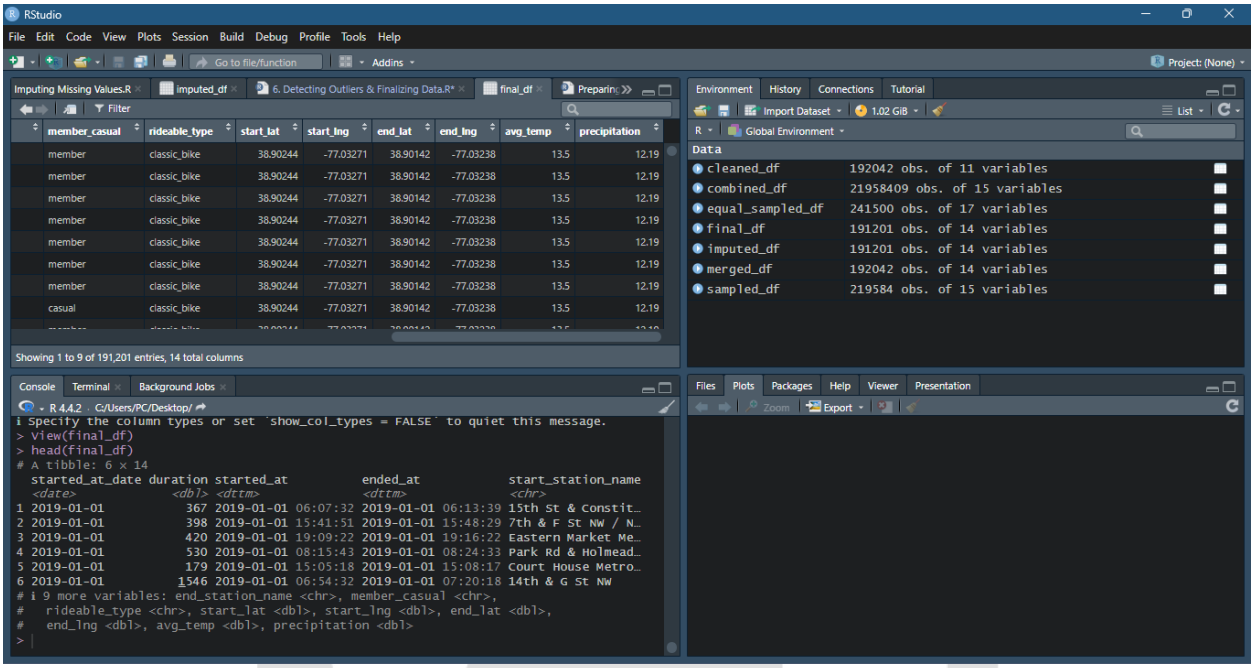


Figure 10. Temperature Boxplot After Removing Outliers

Boxplots for the temperature and precipitation columns show the distribution of data after outliers were removed.



Snapshot 7. Detecting & Handling the Outliers

Comprehensive Data Analysis

Preparing for Data Analysis

First, the summary() function will be applied to the dataframe to extract key summary statistics and other qualitative measures. This will provide an initial understanding of the data patterns and trends related to the bikeshare system prior to analysis.

```
#assigning a new name to the original file
general_sample <- final_df

#extracting summary statistics
summary(general_sample)
```

Code 23. Using summary() on Clean Data

```
 started_at_date       duration        started_at
  Length:191201      Min.   :  20.0   Min.   :2019-01-01 00:34:56.00
  Class :character   1st Qu.: 399.0   1st Qu.:2020-09-09 13:12:44.00
  Mode  :character   Median : 686.0   Median :2022-07-23 18:24:22.00
```

```
                  Mean   : 753.5   Mean   :2022-04-13 20:13:59.77
                  3rd Qu.: 980.0   3rd Qu.:2023-10-25 08:33:39.00
                  Max.   :2384.0   Max.   :2024-10-31 23:46:24.00
   ended_at                    start_station_name end_station_name
Min.   :2019-01-01 00:38:11.0   Length:191201      Length:191201
1st Qu.:2020-09-09 13:26:49.0   Class :character   Class :character
Median :2022-07-23 18:43:09.0   Mode  :character   Mode  :character
Mean   :2022-04-13 20:33:55.7
3rd Qu.:2023-10-25 08:44:25.0
Max.   :2024-10-31 23:58:33.0
member_casual      rideable_type        start_lat       start_lng
Length:191201      Length:191201      Min.   :38.87   Min.   :-77.07
Class :character   Class :character   1st Qu.:38.90   1st Qu.:-77.04
Mode  :character   Mode  :character   Median :38.90   Median :-77.03
                                      Mean   :38.90   Mean   :-77.03
                                      3rd Qu.:38.91   3rd Qu.:-77.02
                                      Max.   :38.93   Max.   :-76.99
   end_lat         end_lng          avg_temp     precipitation
Min.   :38.87   Min.   :-77.08   Min.   :-5.70   Min.   :0.00000
1st Qu.:38.90   1st Qu.:-77.04   1st Qu.:12.30   1st Qu.:0.00000
Median :38.90   Median :-77.03   Median :19.70   Median :0.00000
Mean   :38.90   Mean   :-77.03   Mean   :18.25   Mean   :0.05536
3rd Qu.:38.91   3rd Qu.:-77.02   3rd Qu.:24.90   3rd Qu.:0.00000
Max.   :38.93   Max.   :-76.99   Max.   :33.00   Max.   :1.27000
```

Figure 11. Using summary() on Cleaned Data

Summary of Insights from Capital Bikeshare Dataset Observed through summary() After Data Cleaning

1. **Temporal Patterns:** Trips are likely to peak during specific times of the day (e.g., commuting hours) and seasons (e.g., spring and summer).

2. **Weather Conditions**: Temperatures range from **-5.7°C to 33.0°C**, with most rides occurring in moderate weather (~19°C). While precipitation is minimal for most trips, indicating favorable biking conditions dominate the dataset.

3. **Trip Duration and User Types**: Trips last between **20 seconds and 40 minutes**, with a median duration of ~11 minutes and a mean of ~12.5 minutes, indicating that most trips are short. Longer trips are likely outliers, potentially made by casual riders or for leisure purposes.

4. **Bike Types**: Different bike types, including electric and traditional, are available, which may correlate with user preferences or trip duration.

Quarterly Rentals per Year

This code analyzes usage trends across all four quarters for each year from 2019 to 2024.

```
#for data visualizations
library(tidyverse)
#for managing date and time data
library(lubridate)

#add year and quarter labels by extracting the year from each date
general_sample <- general_sample %>%
  mutate(
    year = year(started_at_date),
    quarter_label = paste0("Q", quarter(started_at_date))
  )

#count the number of rentals for each year-quarter
quarterly_rentals <- general_sample %>%
  group_by(year, quarter_label) %>%
```

```
  summarise(total_rentals = n(), .groups = "drop")

#is quarter_labels being treated as an ordered categorical variable?
#critical for plotting quarters in the correct order
quarterly_rentals <- quarterly_rentals %>%
  mutate(quarter_label = factor(quarter_label, levels = c("Q1", "Q2", "Q3", "Q4")))

#identifies each unique year for the purpose of plotting the data
unique_years <- unique(quarterly_rentals$year)

for (year in unique_years) {
  #filter data for the specific year
  yearly_data <- quarterly_rentals %>% filter(year == !!year)
  #plotting
  plot <- ggplot(yearly_data, aes(x = quarter_label, y = total_rentals, group = 1)) +
    geom_line(color = "red") +
    labs(title = paste("Quarterly Capital Bikeshare Rentals for", year),
         x = "Quarter", y = "Total Rentals") +
    theme_minimal()

  print(plot)
}
```

Code 24. Quarterly Rentals per Year

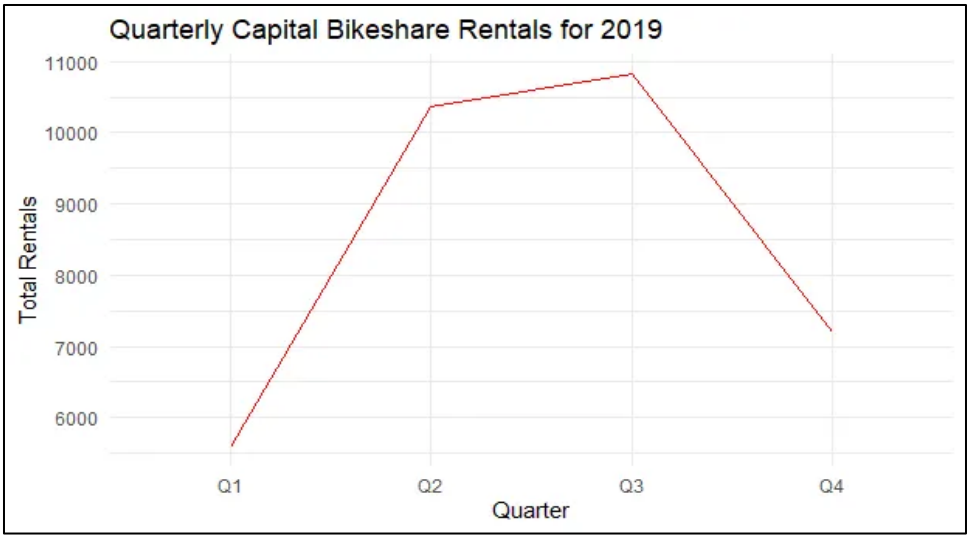


Figure 12. Quarterly Rentals for 2019

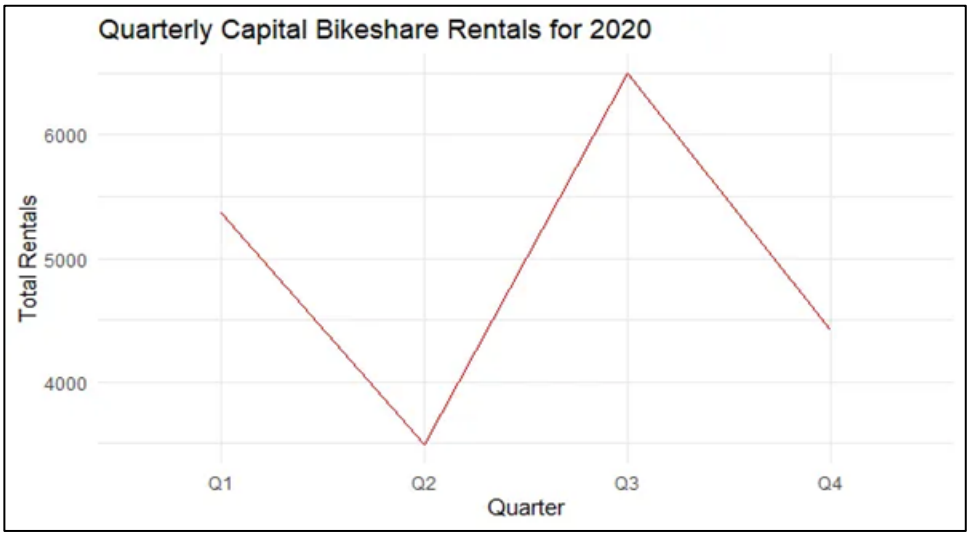


Figure 13. Quarterly Rentals for 2020

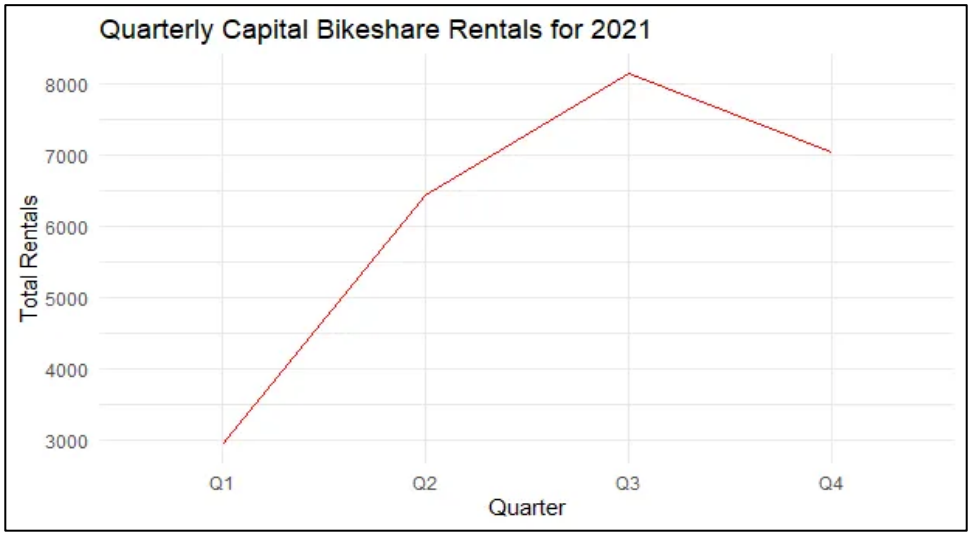


Figure 14. Quarterly Rentals for 2021

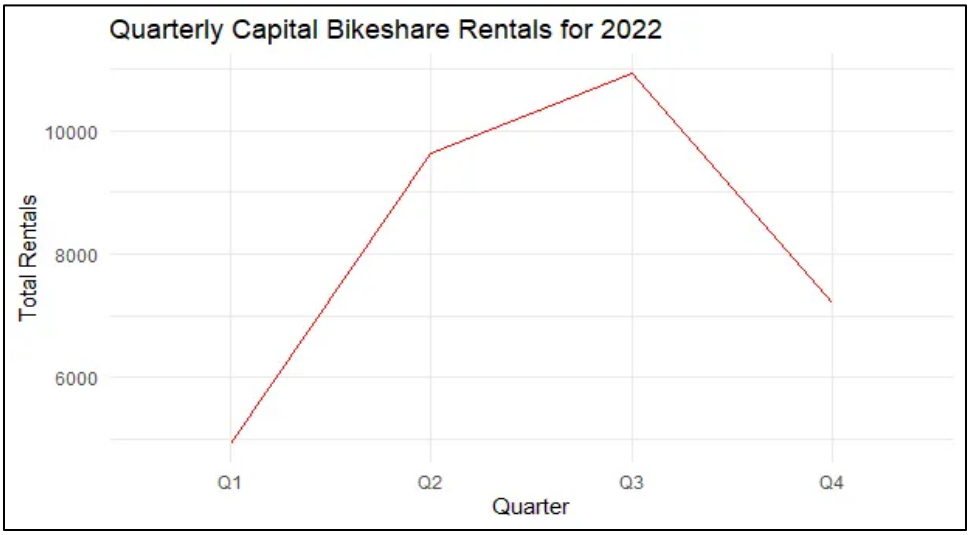


Figure 15. Quarterly Rentals for 2022

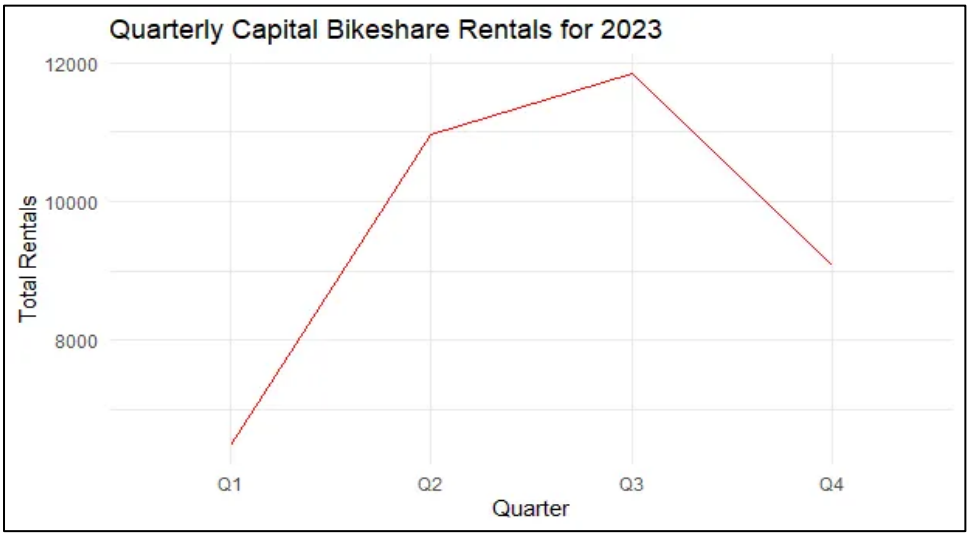


Figure 16. Quarterly Rentals for 2023

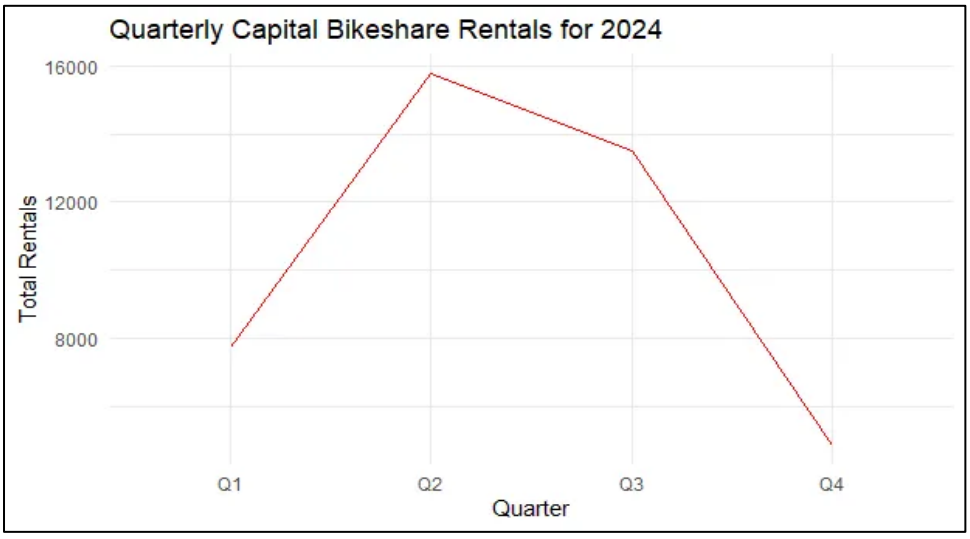


Figure 17. Quarterly Rentals for 2024

Yearly Rental Trends

This code visualizes data about yearly Capital Bikeshare rental trends from 2019 to 2024.

```
library(dplyr)
library(ggplot2)
library(lubridate)

#sample data processing
general_sample <- general_sample %>%
  mutate(year = as.integer(format(started_at_date, "%Y")))

#summarize data yearly
yearly_data <- general_sample %>%
  group_by(year) %>%
  summarise(total_rentals = n())

#plot yearly rental trends
ggplot(yearly_data, aes(x = year, y = total_rentals)) +
  geom_line(color = "red") +
  labs(title = "Yearly Capital Bikeshare Rental Trends from 2019 to 2024", x = "Year", y =
"Total Rentals") +
  theme_minimal()
```

Code 25. Yearly Rental Trends

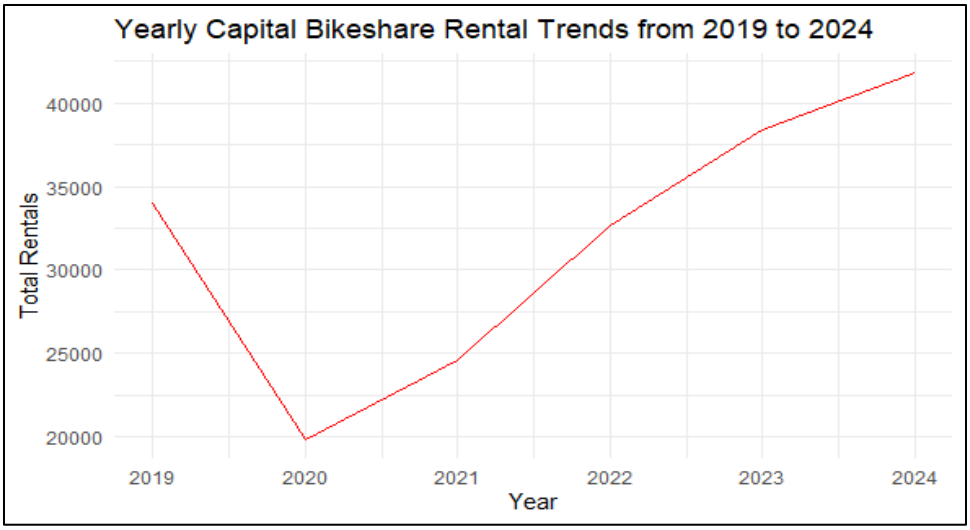


Figure 18. Yearly Rentals from 2019 to 2024

Daily Rental Trends

This analysis aggregates daily rentals into day-of-the-week categories specifically for the second quarter. Q2 was chosen as it often represents spring to early summer in many regions, which is a peak time for outdoor activities like biking, and often have more balanced levels of activity. It helps us focus on a period with likely high usage patterns, avoiding extreme seasonal impacts that could skew results. This method will be applied for all following analyses, as well.

```
library(tidyverse)
library(lubridate)

#extracts year from the date column and adds it as a new column
general_sample <- general_sample %>%
  mutate(year = year(started_at_date))

for (year in unique_years) {
  #filter for the specific year and Q2
  q2_data_year <- general_sample %>%
    filter(year == !!year, quarter_label == "Q2") %>%  # Filter by year and Q2
```

```
    mutate(day_of_week = wday(started_at_date, label = TRUE))  # Add day of the week (e.g., Mon, Tue)

  #combine rentals by day of the week
  weekly_rentals_q2 <- q2_data_year %>%
    group_by(day_of_week) %>%
    summarise(total_rentals = n()) %>%
    arrange(day_of_week)
  #plot
  plot_q2 <- ggplot(weekly_rentals_q2, aes(x = day_of_week, y = total_rentals, fill = total_rentals)) +
    geom_bar(stat = "identity") +
    scale_fill_gradient(low = "#FFCCCC", high = "#990000") +  # Gradient from light red to dark red
    labs(title = paste("Weekly Analysis of Capital Bikeshare Rentals During Q2 for", year),
         x = "Day of the Week", y = "Total Rentals", fill = "Rentals") +
    theme_minimal()

  print(plot_q2)
}
```

Code 26. Daily Rental Trends

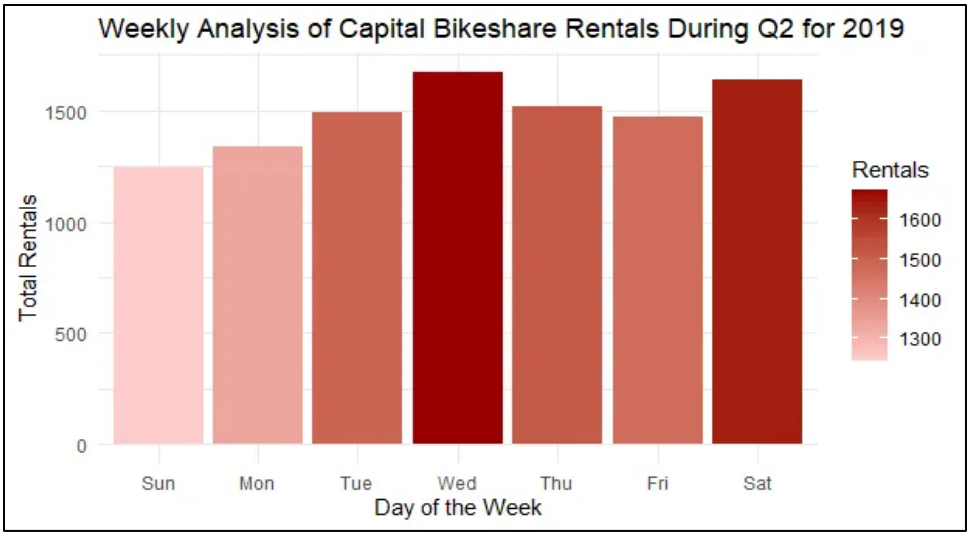


Figure 19. Daily Rentals for 2019

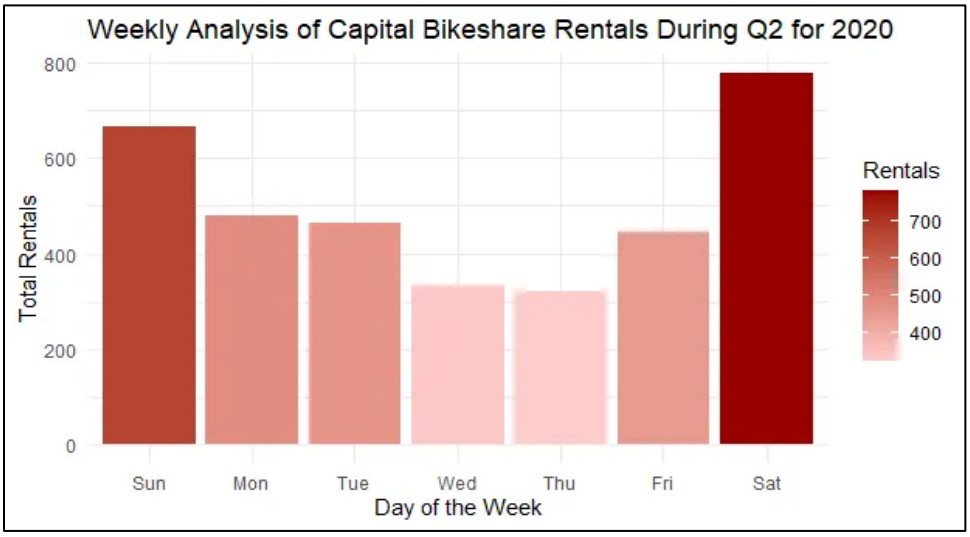


Figure 20. Daily Rentals for 2020

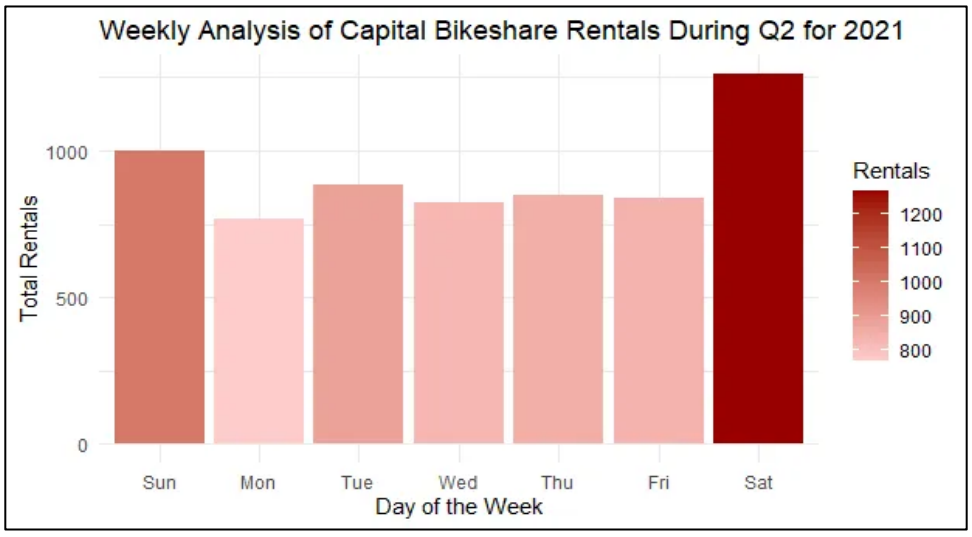


Figure 21. Daily Rentals for 2021

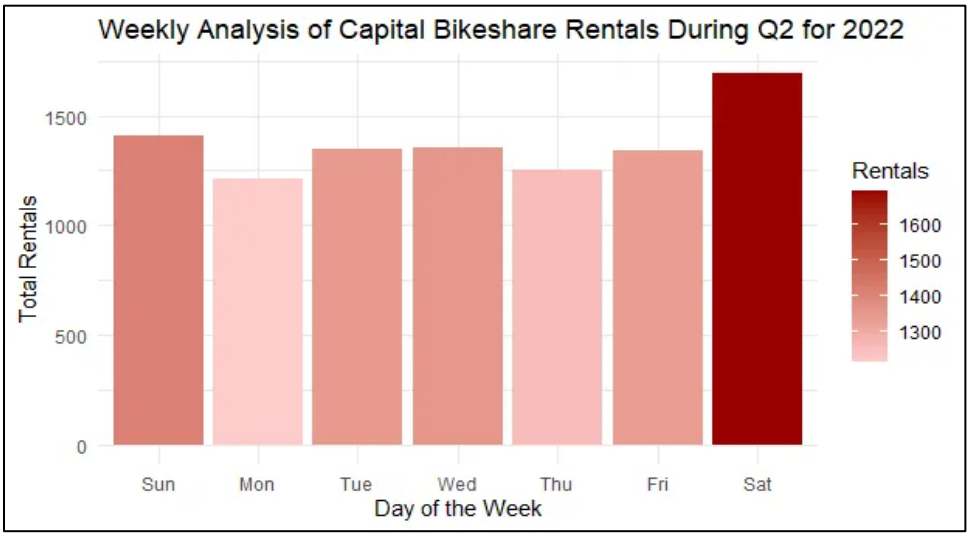


Figure 22. Daily Rentals for 2022

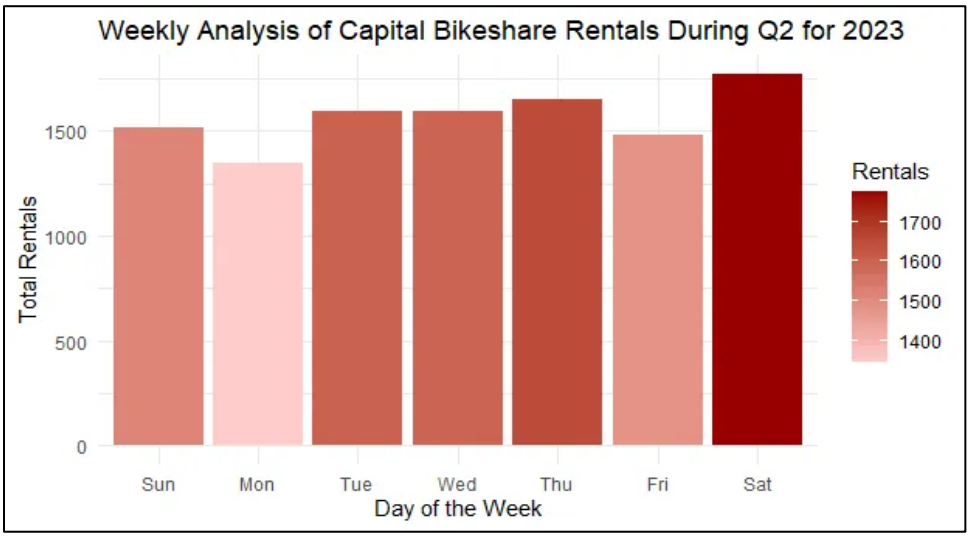


Figure 23. Daily Rentals for 2023

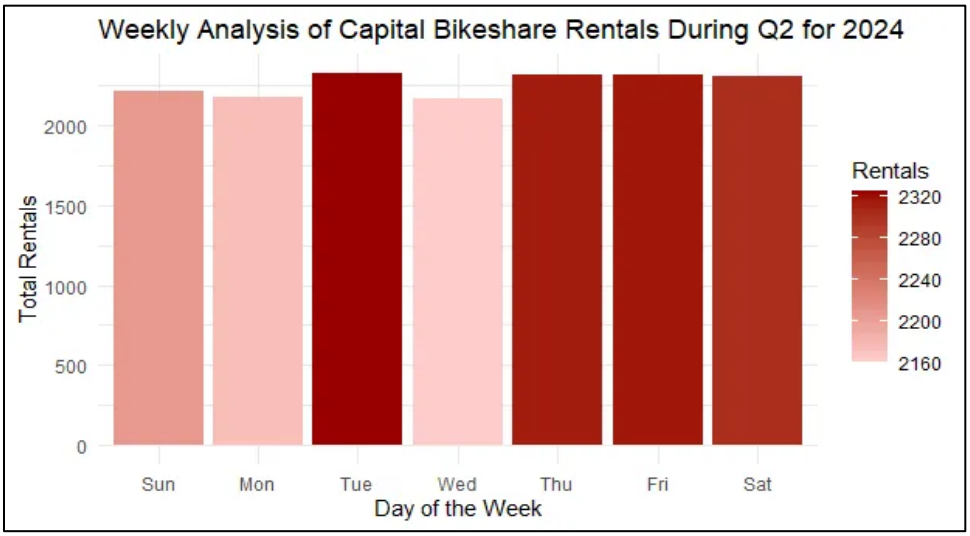


Figure 24. Daily Rentals for 2024

Hourly Rental Trends

This analysis aggregates rentals into hourly categories, ranging from 0:00AM to 23:00PM, specifically for the second quarter.

```r
library(tidyverse)
library(lubridate)

#filter the dataset to include only rows corresponding to Q2
q2_sample <- general_sample %>%
  filter(quarter_label == "Q2")

#extract hour from the started_at column and add to a new column
q2_sample <- q2_sample %>%
  mutate(hour = hour(started_at))

#combine rentals by hour for each year
hourly_rentals <- q2_sample %>%
  group_by(year, hour) %>%
  summarise(total_rentals = n(), .groups = "drop")

#for individual plotting
for (year in unique_years) {
  yearly_data <- hourly_rentals %>% filter(year == !!year)
  #plot
  plot <- ggplot(yearly_data, aes(x = hour, y = total_rentals, fill = hour)) +
    geom_bar(stat = "identity", color = "black") +
    scale_fill_gradient(low = "red", high = "darkred") +
    labs(title = paste("Hourly Bike Rentals During Q2 for", year),
         x = "Hour of the Day", y = "Total Rentals") +
    scale_x_continuous(breaks = 0:23) +  #show all hours from 0 to 23
```

```
    theme_minimal() +
    theme(legend.position = "none")  #remove legend

  print(plot)
}
```

Code 27. Hourly Rental Trends

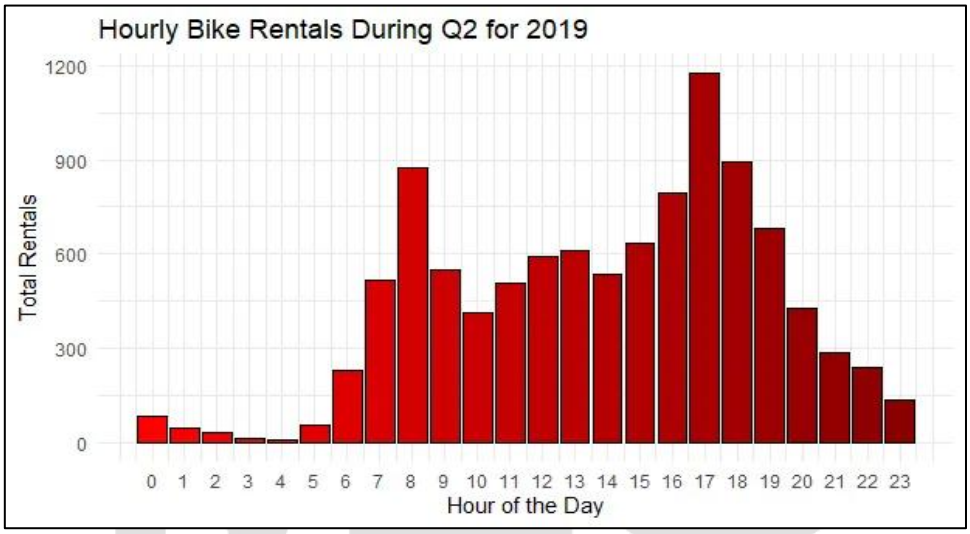


Figure 25. Hourly Rentals for 2019

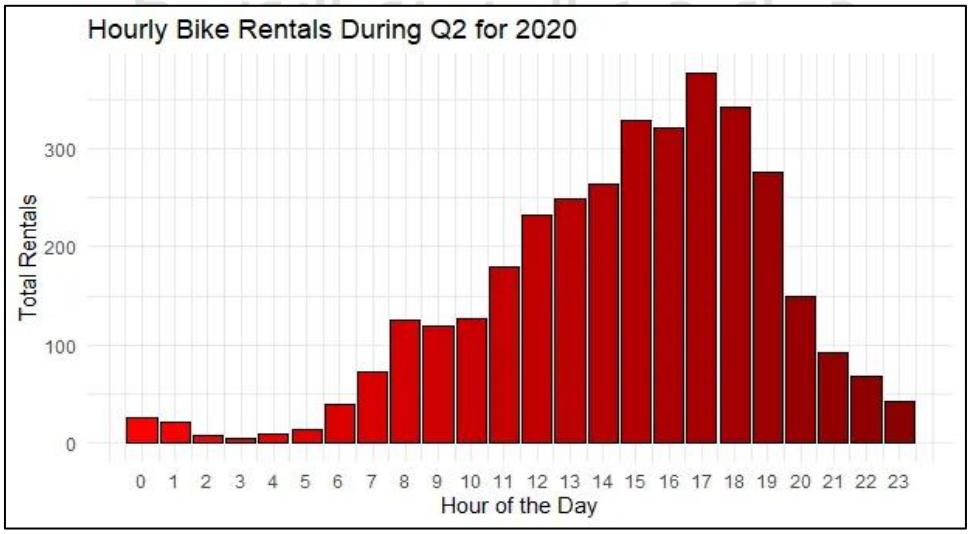


Figure 26. Hourly Rentals for 2020

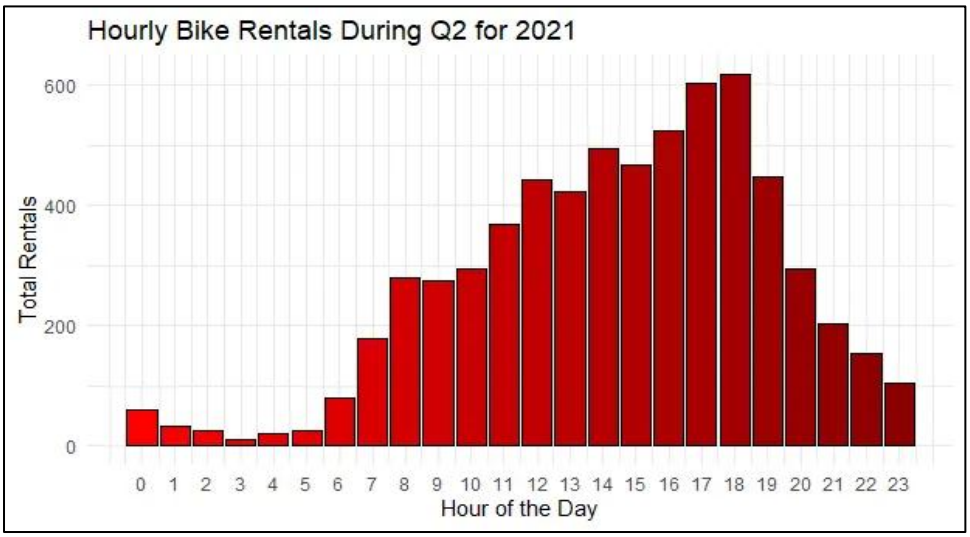


Figure 27. Hourly Rentals for 2021

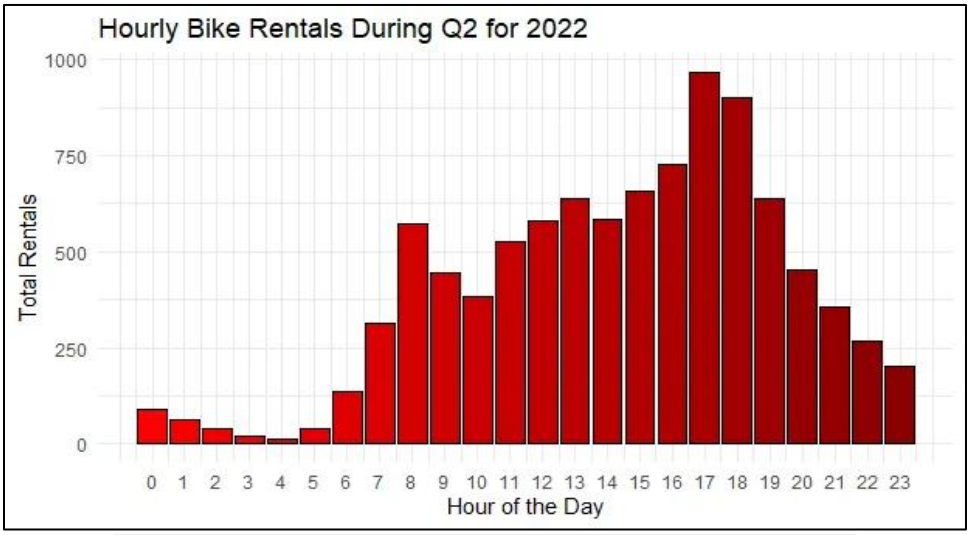


Figure 28. Hourly Rentals for 2022

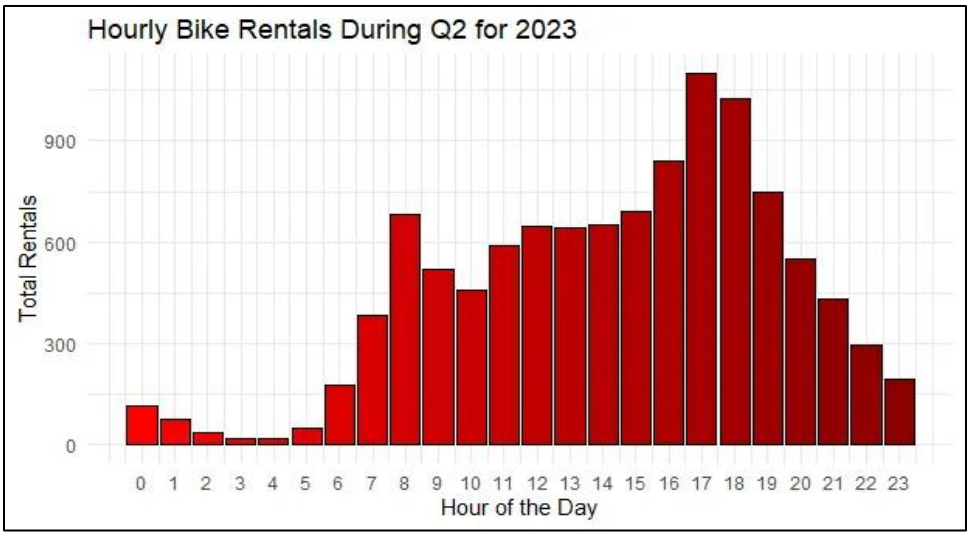


Figure 29. Hourly Rentals for 2023

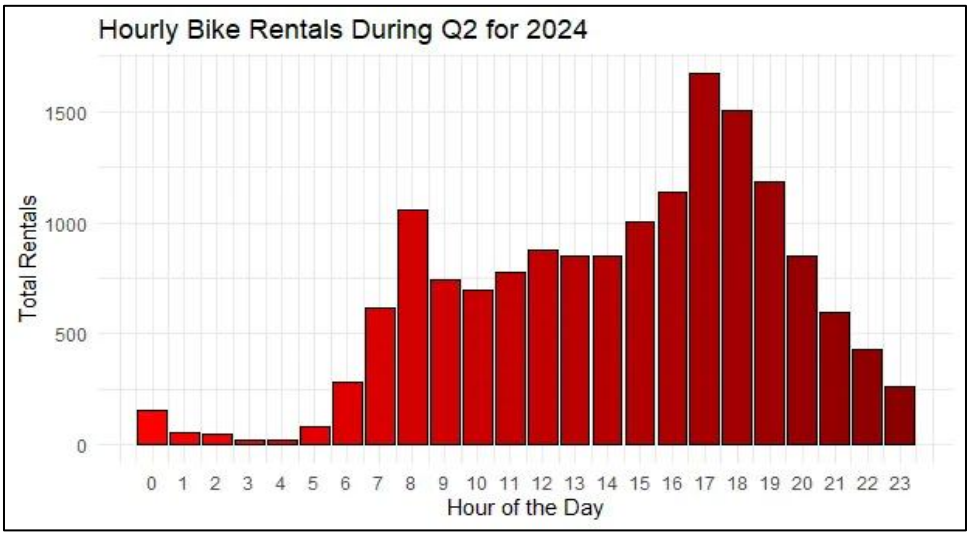


Figure 30. Hourly Rentals for 2024

Top 10 Popular Stations

This code identifies the most popular starting and ending stations based on the number of rentals associated with each.

```
#most popular start stations
popular_start_stations <- general_sample %>%
  group_by(start_station_name) %>%
  summarise(start_count = n()) %>%
  arrange(desc(start_count))

#most popular end stations
popular_end_stations <- general_sample %>%
  group_by(end_station_name) %>%
  summarise(end_count = n()) %>%
  arrange(desc(end_count))

#plot
ggplot(popular_start_stations[1:10,], aes(x = reorder(start_station_name, -start_count), y = start_count)) +
```

```
15.   geom_bar(stat = "identity", fill = "red") +
16.   labs(title = "Top 10 Popular Start Stations", x = "Start Station", y = "Trip Count") +
17.   theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Code 28. Top 10 Popular Stations

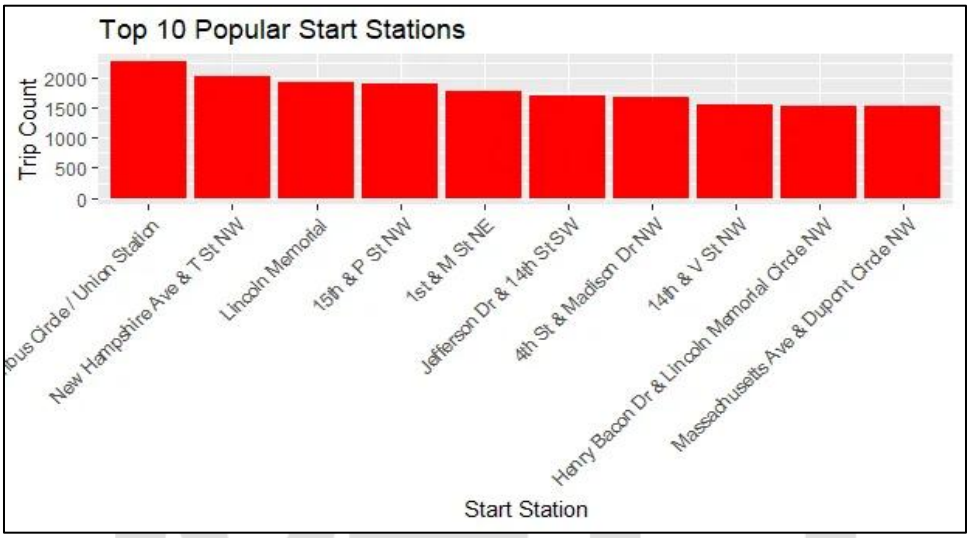


Figure 31. Top 10 Popular Stations

Distribution of Trip Durations by User Type

This code visualizes the distribution of trip durations, and then groups the data by two categories of user types.

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```
1. #histogram for trip durations
2. ggplot(general_sample, aes(x = duration)) +
3.   geom_histogram(binwidth = 300, fill = "red", alpha = 0.7) +
4.   labs(title = "Distribution of Trip Durations", x = "Duration (seconds)", y = "Count")
5.
6. #boxplot for user types
7. ggplot(general_sample, aes(x = member_casual, y = duration, fill = member_casual)) +
8.   geom_boxplot() +
9.   labs(title = "Trip Durations by User Type", x = "User Type", y = "Duration (seconds)")
```

Code 29. Distribution of Trip Durations by User Type

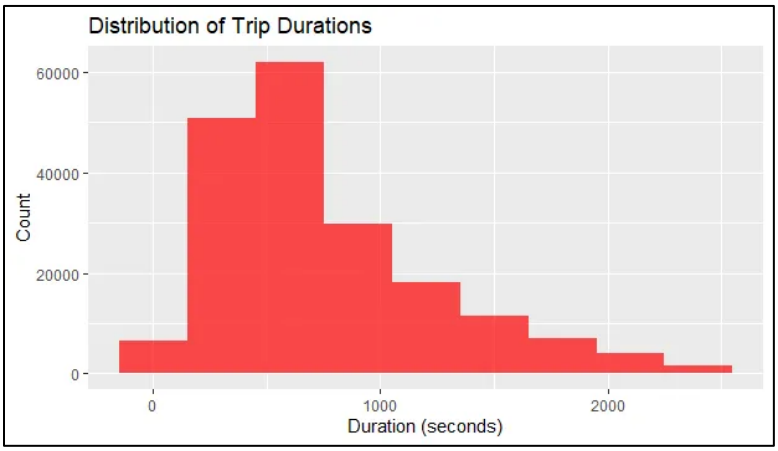


Figure 32. Distribution of Trip Durations

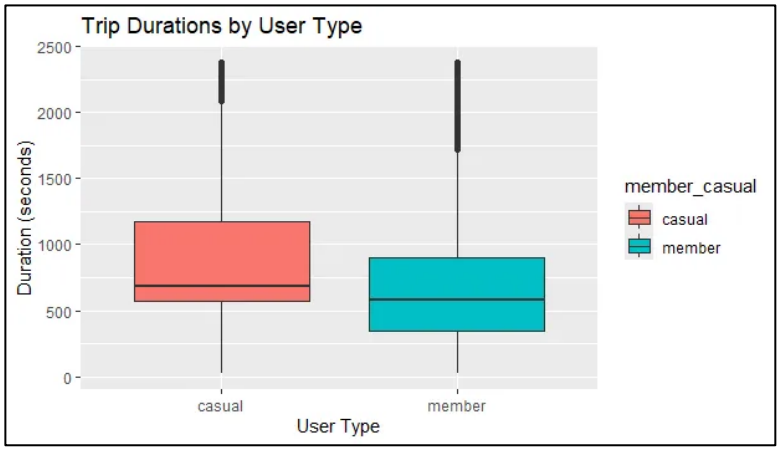


Figure 33. Trip Durations by User Type

Average Trip Durations by Hour for each User Type

This code calculates the average trip duration for each hour of the day, categorized by the two user type.

```
#aggregate trip durations by hour
hourly_analysis <- general_sample %>%
  mutate(hour = lubridate::hour(started_at)) %>%
  group_by(hour, member_casual) %>%
  summarise(avg_duration = mean(duration, na.rm = TRUE))

#plot average duration by hour
ggplot(hourly_analysis, aes(x = hour, y = avg_duration, color = member_casual)) +
  geom_line(size = 1) +
  labs(title = "Average Trip Duration by Hour", x = "Hour of Day", y = "Average Duration (seconds)")
```

Code 30. Average Trip Durations by Hour for Each User Type

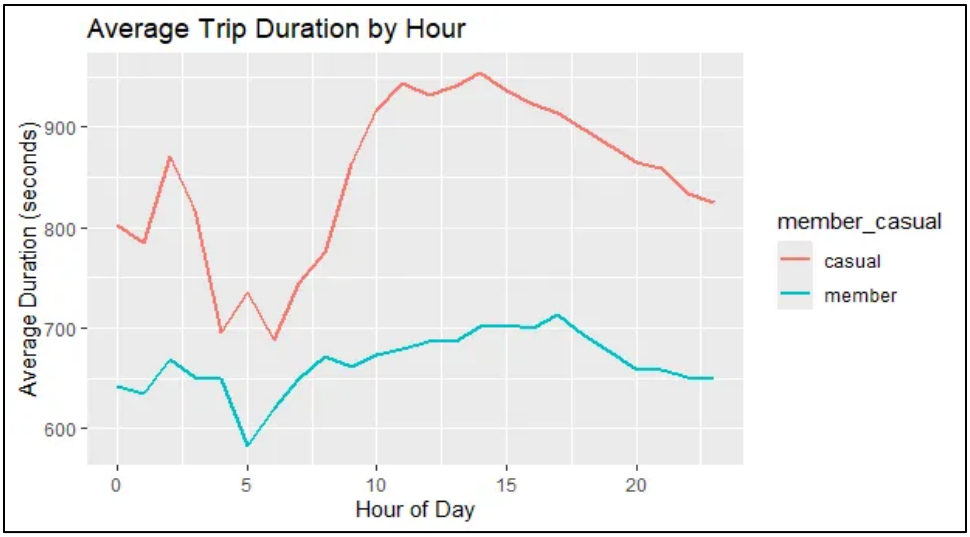


Figure 34. Average Trip Duration by Hour

Weather Correlation

This code performs an analysis to explore the relationships between average temperature, total bike rentals, and average trip durations, using aggregated quarterly data.

```
#for summarization
library(dplyr)
#for visualization
library(ggplot2)

#combine data by quarters
quarterly_data <- general_sample %>%
  group_by(quarter_label) %>%
  summarise(
    total_rentals = n(),
    avg_duration = mean(duration, na.rm = TRUE),
    avg_temp = mean(avg_temp, na.rm = TRUE),
  )

#correlations
```

```
cor_temp_rentals <- cor(quarterly_data$avg_temp, quarterly_data$total_rentals, use = "complete.obs")
cor_temp_duration <- cor(quarterly_data$avg_temp, quarterly_data$avg_duration, use = "complete.obs")

#print correlations
cat("Correlation between temperature and total rentals:", cor_temp_rentals, "\n")
cat("Correlation between temperature and average duration:", cor_temp_duration, "\n")

#linear models
model_rentals <- lm(total_rentals ~ avg_temp, data = quarterly_data)
model_duration <- lm(avg_duration ~ avg_temp, data = quarterly_data)

#model summaries
cat("\nModel for Total Rentals:\n")
summary(model_rentals)

cat("\nModel for Average Duration:\n")
summary(model_duration)

#visualizing temperature vs. total rentals
ggplot(quarterly_data, aes(x = avg_temp, y = total_rentals)) +
  geom_point() +
  geom_smooth(method = "lm", color = "red") +
  labs(
    title = "Temperature vs. Total Rentals",
    x = "Average Temperature (°C)",
    y = "Total Rentals"
  )

#visualizing temperature vs. average duration
```

```r
ggplot(quarterly_data, aes(x = avg_temp, y = avg_duration)) +
  geom_point() +
  geom_smooth(method = "lm", color = "red") +
  labs(
    title = "Temperature vs. Average Duration",
    x = "Average Temperature (°C)",
    y = "Average Duration (s)"
  )
```

Code 31. Weather Correlation

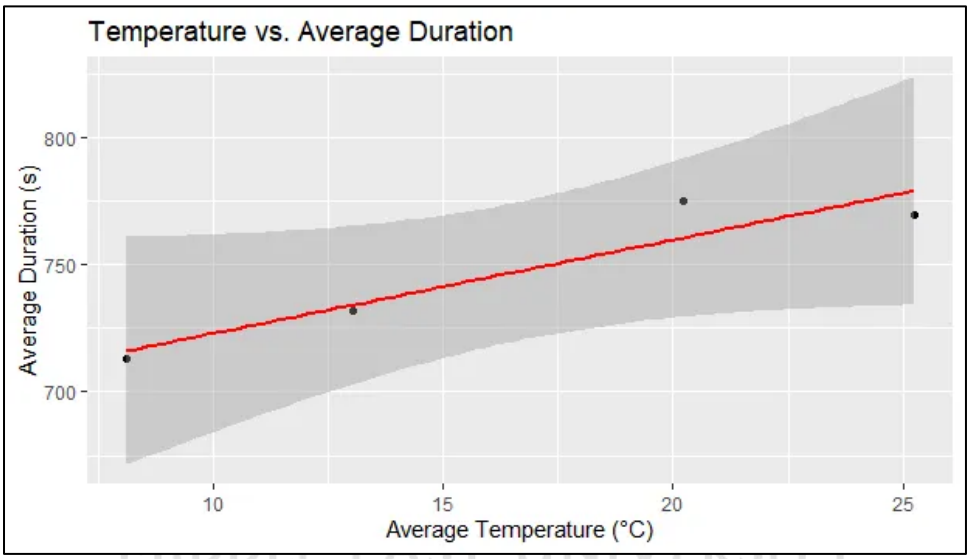


Figure 35. Temperature vs. Average Duration

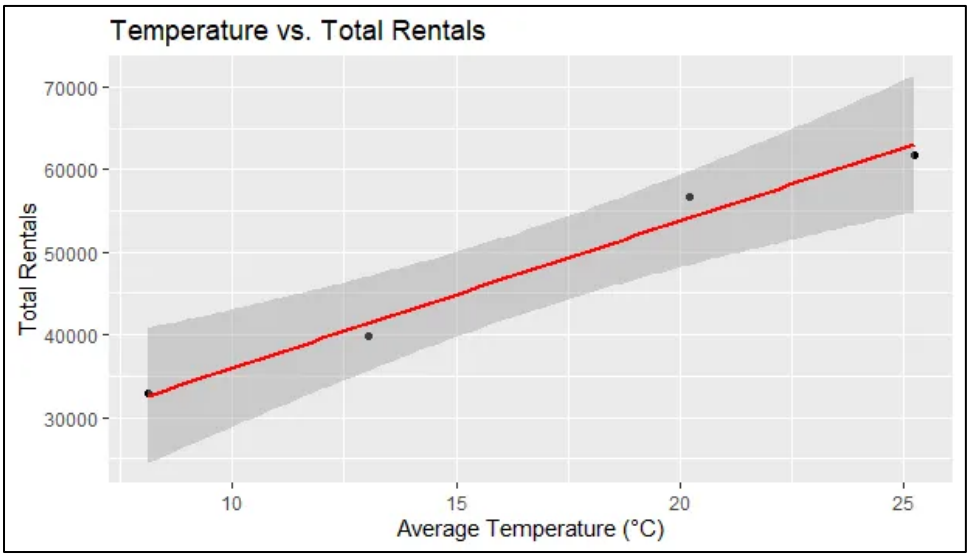


Figure 36. Temperature vs. Total Rentals

Time-Series Forecasting

This section uses a time series approach to create forecasts for bike rentals. An ARIMA (Autoregressive Integrated Moving Average) model is fitted to the time series data using the auto.arima() function, which automatically selects the best ARIMA model parameters based on the data. The model's accuracy is evaluated using the Mean Absolute Percentage Error (MAPE), which measures the average percentage deviation between the predicted and actual values.

```
library(forecast)

#daily rentals to monthly rentals
daily_rentals <- as.data.frame(table(general_sample$started_at_date))
colnames(daily_rentals) <- c("date", "total_rentals")
daily_rentals$date <- as.Date(daily_rentals$date)
daily_rentals$year_month <- format(daily_rentals$date, "%Y-%m")

#aggregation
monthly_rentals <- aggregate(total_rentals ~ year_month, data = daily_rentals, sum)
colnames(monthly_rentals) <- c("month", "total_rentals")

monthly_rentals$month <- as.Date(paste0(monthly_rentals$month, "-01"))

#create time series object
months.ts <- ts(monthly_rentals$total_rentals,
                start = c(as.numeric(format(min(monthly_rentals$month), "%Y")),
                          as.numeric(format(min(monthly_rentals$month), "%m"))),
                frequency = 12)

#decompose the time series
```

```r
decomp <- decompose(months.ts)

#forecast for 5 years
forecast_horizon <- 12 * 5

#fit arima model on the entire dataset
fit <- auto.arima(months.ts)

#generate 5-year forecasts
forecast_values <- forecast(fit, h = forecast_horizon)

#split the data for mape calculation, using the last 12 months for testing
train_size <- length(months.ts) - 12
train_ts <- window(months.ts, end = c(as.numeric(format(monthly_rentals$month[train_size], "%Y")),
                                      as.numeric(format(monthly_rentals$month[train_size], "%m"))))
test_ts <- window(months.ts, start = c(as.numeric(format(monthly_rentals$month[train_size + 1], "%Y")),
                                       as.numeric(format(monthly_rentals$month[train_size + 1], "%m"))))

#fit arima on training data only
fit_train <- auto.arima(train_ts)

#forecast for the test period to calculate mape
test_forecast <- forecast(fit_train, h = length(test_ts))

#calculate mape
mape <- mean(abs((test_ts - test_forecast$mean) / test_ts)) * 100
cat("mape:", round(mape, 2), "%\n")
```

```
49.
50. #plot the 5-year forecast
51. plot(forecast_values, main = "5-year Forecast of Bike Rentals")
52. lines(months.ts, col = 2, lwd = 2)  #actual data in red
53.
```

Code 32. Time-Series Forecasting

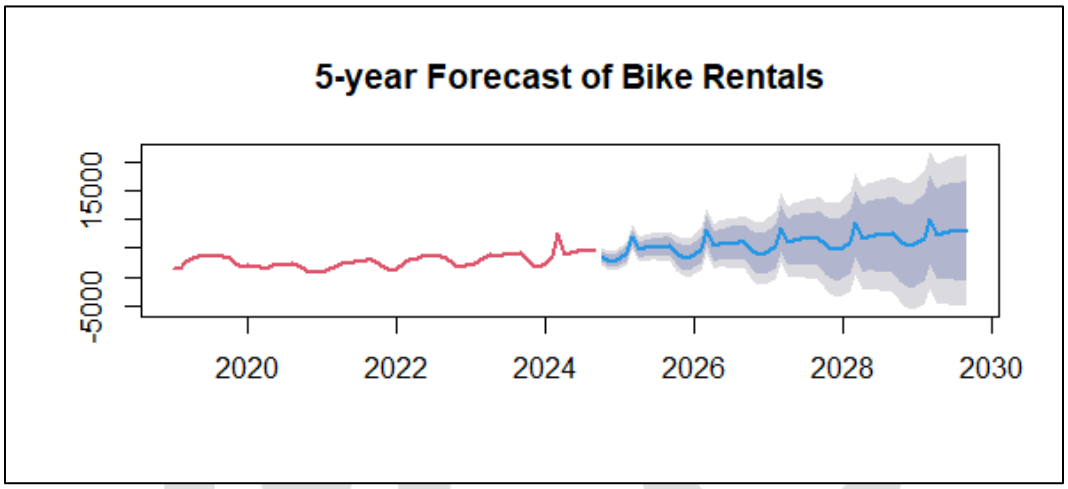


Figure 37. Forecast of Bike Rentals Across Multiple Years

The forecasting model achieved a Mean Absolute Percentage Error (MAPE) of 10.87%, indicating a reasonably accurate prediction, with an average error of approximately 10.87% when compared to the actual values.

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Interactive Dashboard

An interactive dashboard is an important component for the bike-sharing system project, acting as a user-friendly platform for visualizing and understanding complex data. intended to visualize complex data in a user-friendly way for public purposes. These dashboards, developed using Tableau and Power BI, can be undertaken by stakeholders to understand how the bike-sharing system operates and explore trends. It is intended to have a very friendly approach so that anyone is capable of filtering and drilling down into data to arrive at patterns and factors behind rentals.

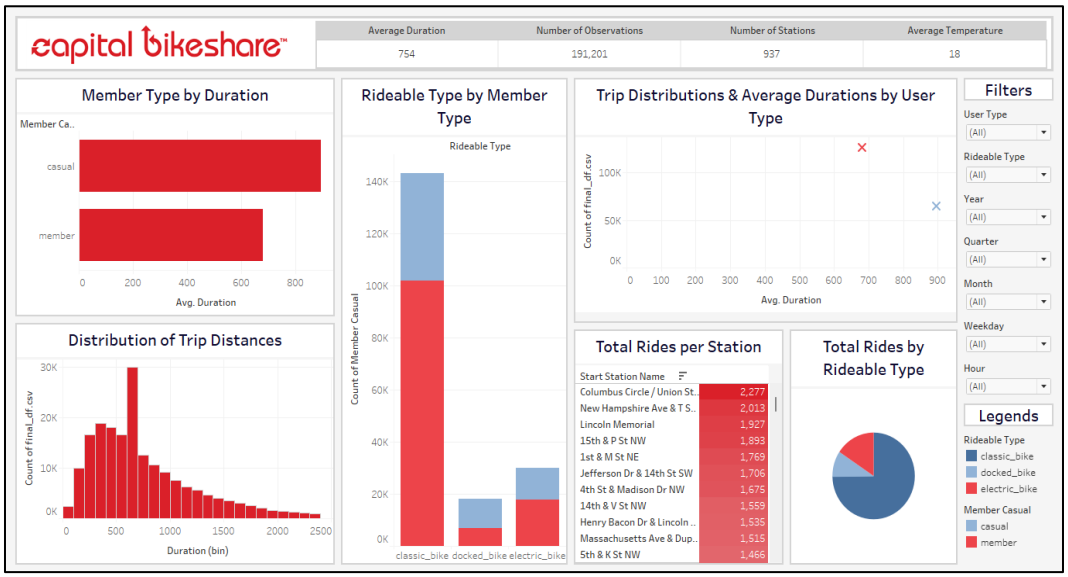


Figure 38. Tableau Dashboard Page One

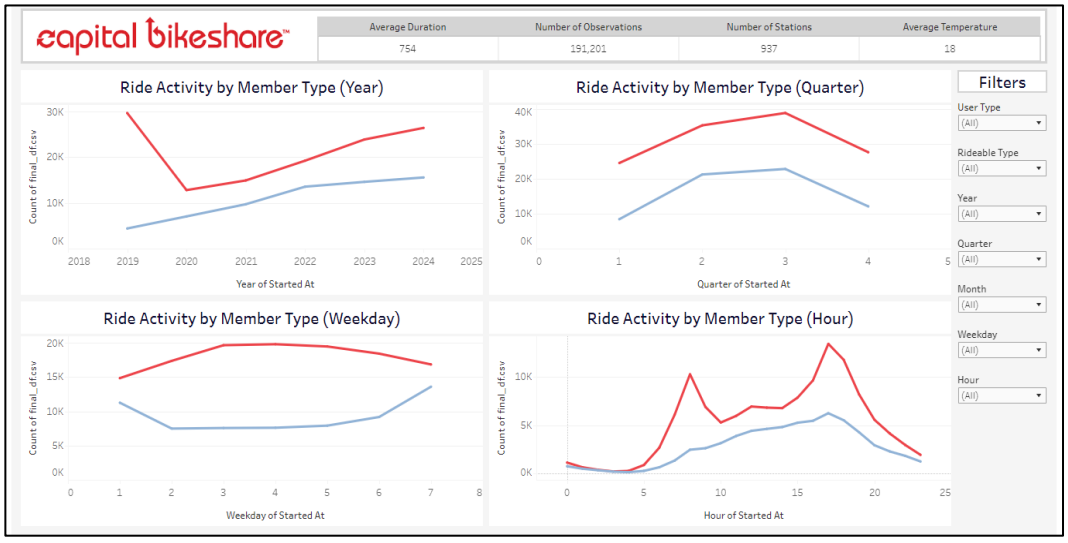


Figure 39. Tableau Dashboard Page Two

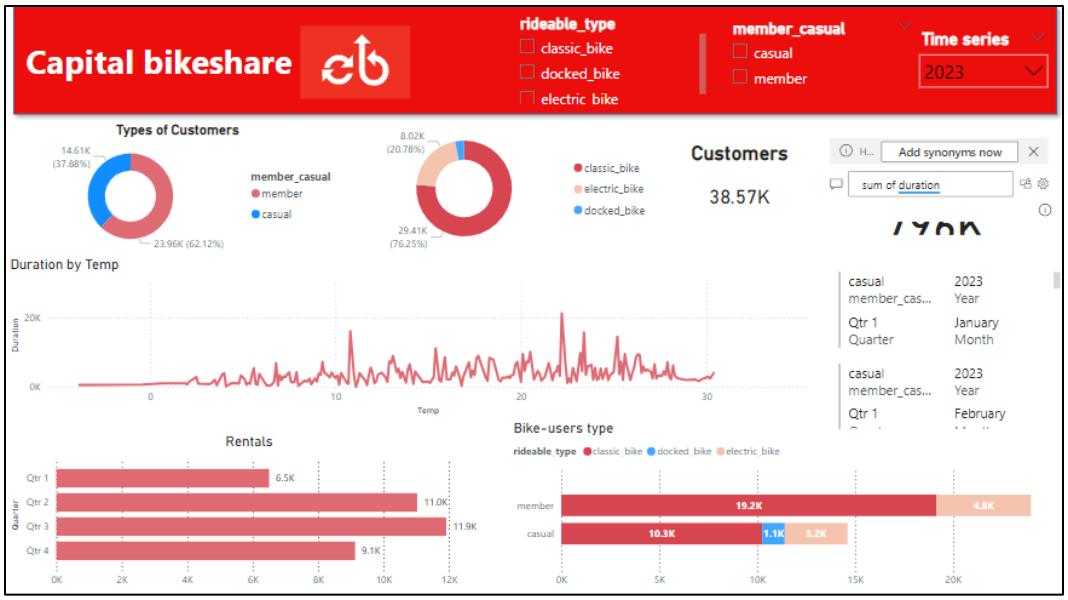


Figure 40. Power BI Dashboard

Components of the Dashboard

Header

1. **Logo:** The Capital Bikeshare logo is prominently displayed in the top left corner.
2. **Scorecards:** A bird's eye view of key variables such as average trip duration, rides, station counts, and average temperature.

Filters

The filters allow users to further fine-tune the information being viewed in the dashboard such as type of user, type of rideable, year, quarter, month, weekday, and even hour.

Legends

The legends at the bottom-right corner act as guides that explain the meaning of the colors for rideable and user types.

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Visualizations

1. **Member Type by Duration:** A bar graph comparing the average trip duration by casual users and members.
2. **Rideable Type by Member:** A 100 % stacked bar showing how each rideable type is used among casual users and members.
3. **Trip Distributions & Average Durations by User Type:** A scatter plot indicating the relationship of trip duration on another distribution categorized by user type.
4. **Distribution of Trip Distances:** A histogram concerning the duration of trips.
5. **Total Rides per Station:** A highlighted table that compares various station rental counts and their intensity of color to determine the most popular stations compared to the rest.
6. **Total Rides by Rideable Type:** A pie chart that explains the overall portion of rides taken for every bike type.
7. **Ride Activity by Member Type (Year):** A chart that depicts the changing extent of rides used by casuals compared to members over the years from 2019 to 2024.
8. **Ride Activity by Member Type (Quarter):** This one accommodates ride activity by user types for every quarter of the year.
9. **Ride Activity by Member Type (Weekday):** Breakdowns of ride activity by day of the week are shown in this visualization.
10. **Ride Activity by Member Type (Hour):** A ride activity chart at an hourly rate throughout the day.

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Results & Findings

The analysis of the Capital Bikeshare system's usage patterns from 2019 to 2024 reveals several insights. These findings are derived from a comprehensive data analysis, which includes descriptive statistics, temporal trend analysis, weather correlation analysis, predictive analytics, and so on.

Peak Usage Times and Seasonal Variation

The analysis of Capital Bikeshare data highlights distinct temporal patterns in bike rentals, influenced by daily, weekly, and seasonal factors:

1. **Commuting Hours:** A marked increase in bike rentals during typical commuting hours, particularly on weekdays, suggests that many users rely on the service for work or school commutes.
2. **Seasonal Weather:** Warmer months consistently show higher rental volumes, while colder months and rainy days correspond with reduced usage, emphasizing the impact of weather conditions on cycling behavior.
3. **Leisure and Tourism:** Weekend usage patterns differ from weekdays, with higher daytime rentals indicative of leisure and tourism activities.
4. **Quarterly Trends:** A seasonal variation is evident across quarters, with the second quarter (spring and early summer) experiencing the highest rental volumes.

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Station Popularity

Certain stations experienced higher usage than others, likely due to their proximity to areas with greater demand, such as commercial or recreational zones.

Impact of Weather

The analysis reveals a positive correlation between temperature and bike rentals. Warmer weather promotes higher rental volumes and longer trip durations. Conversely, colder temperatures deter bike usage.

User Behavior

The majority of trips across all users are short, supporting the use of bikes as a convenient mode of transportation for short-distance travel within the city. Distinct patterns emerge between casual users and members:

1. **Members:** Primarily utilize bikes for short commutes, with trips often concentrated during peak commuting hours.
2. **Casual Users:** Rent bikes for leisure or tourism, often taking longer trips.

Forecasting Future Rentals

The red portion of the line represents historical data. It shows some cyclical fluctuations in bike rentals, but overall, it appears relatively stable. The section of the line in blue represents the forecast. It shows a clear upward trend, indicating an expected increase in bike rentals over the coming years. The shaded areas around the forecasted line represent confidence intervals. These intervals show the range within which the actual number of bike rentals is likely to fall.

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Discussion

Analyzing the Capital Bikeshare system's usage patterns gives greater insight into its role in urban mobility, user behavior, and other external factors such as weather and seasonality. Such analysis can suggest various improvement and expansion opportunities for the system through interpretation of results. The discussion points to key findings, implications, and actionable recommendations for optimizing service for both operators and users. Key insights to formulate strategies designed to meet different user needs while solving complicated operational problems in a very dynamic urban environment.

Temporal and Seasonal Trends

The observed peak usage during commuting hours underscores the system's role in urban transportation. The seasonal variation, with higher rentals during warmer months, aligns with expectations, as weather significantly influences outdoor activities. This suggests the need for strategic fleet allocation to meet increased demand during peak seasons and times.

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Station Popularity

Variations in station usage highlight the importance of strategic station placement. Expanding the system to underserved areas or increasing bike availability in high-demand zones could enhance overall accessibility and user satisfaction.

Impact of Weather

The positive correlation between temperature and bike rentals can be explained by the fact that people are more comfortable cycling in warmer conditions. On the other hand, colder temperatures discourage cycling due to discomfort and safety concerns. There is also a positive

correlation between temperature and average trip duration, indicating that people take longer rides when the weather is more pleasant.

User Behavior

The study also considers the difference between casual users and members. Long-term users (members) are more likely to rent bikes for commuting while short-term users may rent bikes for leisure or tourism. This difference in user motivation and needs leads to different usage patterns. The data also shows that most trips are short, suggesting that many people use bikes for short-distance travel within the city. Longer trips may be taken by those using bikes for recreational purposes.

Recommendations for System Improvement

To enhance the overall efficiency, user satisfaction, and sustainability of the Capital Bikeshare system, several targeted recommendations for service improvement have been proposed. Expanding station coverage in high-demand areas, such as residential neighborhoods, business districts, and tourist attractions, would increase accessibility. Introducing flexible docking systems, including virtual docks using GPS technology, could reduce reliance on physical stations. To address weather challenges, seasonal offers and promotions during colder months could encourage usage, while all-weather bikes equipped with mudguards, anti-slip tires, and lights would ensure safety in varying conditions. Additionally, developing underground bike lanes where subway systems are in place and installing clear domes or canopies over bike lanes would provide year-round protection from harsh weather, promoting continuous usage. For commuter incentives, offering loyalty programs, discounted memberships, and integrating the service with public transport networks through combined travel passes and co-located stations would attract long-term users. Mobile app improvements, including real-time availability,

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personalized trip recommendations, and gamification features, could further enhance the user experience, along with 24/7 customer support for prompt issue resolution. To ensure long-term sustainability, expanding the availability of e-bikes and equipping docking stations with solar panels for renewable energy would reduce environmental impact and attract a broader demographic.

Recommendations for Implementing a Bike-Sharing System in Jordan

To successfully implement a bike-sharing system in Jordan, specific recommendations tailored to the country's unique transportation needs and urban landscape are essential. First, bike stations should be prioritized in high-traffic areas, near public transport hubs, educational institutions, and tourist attractions, with an emphasis on locations with flatter terrain that are more suitable for bike riding, given that Jordan's landscape features many hilly areas. Additionally, affordable pricing should be offered to make the system accessible across different socioeconomic groups, alongside the inclusion of e-bikes for more hilly areas. Promoting physical activity through community awareness campaigns highlighting the health benefits of cycling and providing incentives for active users, such as discounts or rewards for frequent trips, would further encourage adoption. Addressing Jordan’s hot climate, shaded rest stops and hydration stations along popular cycling routes are necessary, along with developing dedicated bike lanes and parking zones to ensure rider safety. Collaboration with tourism agencies to offer guided cycling tours around historical and scenic sites, as well as ensuring multilingual support for mobile apps and instructions, would integrate the system with Jordan’s tourism sector. To support sustainability, local partnerships with businesses to sponsor stations and the use of eco-friendly bikes would be important, while community involvement in maintaining the system would foster a sense of ownership and contribute to its long-term success.

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Conclusion

This study provides a comprehensive analysis of the Capital Bikeshare system's usage patterns from 2019 to 2024, uncovering critical insights into how temporal trends, weather conditions, and user behavior influence bike rental activity. The findings reveal that bike-sharing services experience significant variation in usage based on commuting hours, seasonal changes, and weather patterns. Peak usage during weekday commuting hours, higher rentals in warmer months, and the reduced frequency of use during colder or rainy periods emphasize the role of external factors in determining demand. The behavior of casual users and members also differs, with casual users favoring leisure activities and members primarily using bikes for commuting, further diversifying the system's usage patterns.

To address these challenges and maximize the system's potential, a series of recommendations for service improvement has been proposed. These include expanding station coverage in high-demand areas, integrating flexible docking options, enhancing bike infrastructure with weather-resilient features such as clear domes, and offering incentives to commuters through loyalty programs and integration with public transportation. Sustainability initiatives, such as increasing the availability of e-bikes and utilizing renewable energy-powered stations, aim to make the system more environmentally friendly while accommodating a wider demographic. By addressing these areas, the Capital Bikeshare system can enhance its accessibility, user satisfaction, and operational efficiency, ensuring its growth and long-term sustainability.

The exploration of implementing a bike-sharing system in Jordan extends the scope of this study to new markets. Recommendations tailored to Jordan's unique urban landscape and climate include prioritizing high-traffic areas, addressing affordability to cater to diverse

socioeconomic groups, and providing shaded rest stops and hydration stations to adapt to the country's hot climate. Additional measures, such as running community campaigns to promote cycling and collaborating with local tourism agencies to integrate bike-sharing with tourism initiatives, aim to foster a cycling culture while supporting sustainable urban mobility.

In conclusion, bike-sharing systems like Capital Bikeshare offer significant potential to improve urban mobility, promote public health, and contribute to sustainability goals. This study highlights the importance of data-driven decision-making to optimize existing services and adapt them to new environments. By addressing user needs, environmental conditions, and operational challenges, bike-sharing systems can play a pivotal role in shaping smarter, more sustainable cities worldwide.

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Appendix

Appendix A: Raw Data Description

Dataset Name: Capital Bikeshare Usage Data (2019–2024)

Description: The dataset comprises detailed information about daily bike rental records, including station information, and user types.

Column Name	Description
ride_id	Unique identifier for each bike rental trip
started_at	Timestamp of when the trip began
ended_at	Timestamp of when the trip ended
start_station_name	Name of the station where the trip began
end_station_name	Name of the station where the trip ended
start_lat	Latitude of the start station location
start_lng	Longitude of the start station location
end_lat	Latitude of the end station location
end_lng	Longitude of the end station location
user_type	Type of user (e.g., member, casual)

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Appendix B: Data Tables

This section provides the data tables used throughout the project. Each table represents a key step in the analysis process, including data collection, cleaning, transformation, and any intermediate datasets generated for analysis.

Data Tables Overview

Below are the data tables used in different stages of the project:

1. **Sampled Data**: A subset of the original data, used for exploratory analysis or creating visualizations.

duration	started_at	ended_at	start_station_id	start_station_name	end_station_id	end_station_name	bike_number	member_casual	ride_id	rideable_type	start_lat	start_lng	end_lat	end_lng
230	2019-01-01 00:04:48	2019-01-01 00:08:39	31,203	14th & Rhode Island Ave NW	31,200	Massachusetts Ave & Dupont Circle NW	E00141	Member						
1,549	2019-01-01 00:06:37	2019-01-01 00:32:27	31,321	15th St & Constitution Ave NW	31,114	18th St & Wyoming Ave NW	W24067	Casual						
177	2019-01-01 00:08:46	2019-01-01 00:11:44	31,104	Adams Mill & Columbia Rd NW	31,323	Woodley Park Metro / Calvert St & Connecticut Ave NW	W22654	Casual						
228	2019-01-01 00:08:47	2019-01-01 00:12:35	31,281	8th & O St NW	31,280	11th & S St NW	W22336	Member						
1,300	2019-01-01 00:12:29	2019-01-01 00:34:10	31,014	Lynn & 19th St North	31,923	Columbia Pike & S Taylor St	70004	Member						

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2. **Cleaned Data**: The table resulting from the data cleaning process, where missing values were handled, and irrelevant data points were removed. As well as data transformations made for the purpose of analysis, such as calculating new variables, aggregating data, or reshaping the data.

started_at_date	duration	started_at	ended_at	start_station_name	end_station_name	member_casual	rideable_type	start_lat	start_lng	end_lat	end_lng	avg_temp	precipitation
2019-01-01	367	2019-01-01 06:07:32	2019-01-01 06:13:39	15th St & Constitution Ave NW	6th St & Indiana Ave NW	member	classic_bike	38.90244	-77.03271	38.90142	-77.03238	13.5	12.19
2019-01-01	398	2019-01-01 15:41:51	2019-01-01 15:48:29	7th & F St NW / National Portrait Gallery	13th St & New York Ave NW	member	classic_bike	38.90244	-77.03271	38.90142	-77.03238	13.5	12.19
2019-01-01	420	2019-01-01 19:09:22	2019-01-01 19:16:22	Eastern Market Metro / Pennsylvania Ave & 7th St SE	Lincoln Park / 13th & East Capitol St NE	member	classic_bike	38.90244	-77.03271	38.90142	-77.03238	13.5	12.19
2019-01-01	530	2019-01-01 08:15:43	2019-01-01 08:24:33	Park Rd & Holmead Pl NW	Adams Mill & Columbia Rd NW	member	classic_bike	38.90244	-77.03271	38.90142	-77.03238	13.5	12.19
2019-01-01	179	2019-01-01 15:05:18	2019-01-01 15:08:17	Court House Metro / 15th & N Uhle St	Wilson Blvd & Franklin Rd	member	classic_bike	38.90244	-77.03271	38.90142	-77.03238	13.5	12.19

Code Used for Data Table Creation

Below is an example of creating a table from the dataset using flextable:

```
1. library(flextable)
2.
3. #only select the first 5 rows and keeping all columns
4. sampled_df_small <- final_df[1:5, ]
5.
6. flextable(sampled_df_small)
```

Appendix C: Limitations of the Study

The analysis faced several constraints, which may impact the generalizability of the findings:

1. **Data Completeness:** The dataset lacked data on bike maintenance schedules, which could influence station availability.
2. **Weather Data Limitations:** Weather conditions were averaged and may not capture microclimatic variations that affect bike rentals.
3. **Temporal Scope:** The analysis spanned from 2019 to 2024, but data gaps for specific periods could limit the accuracy of trends.
4. **Lack of Demographic Data:** Insights into rider demographics (e.g., age, gender) were unavailable, restricting deeper behavioral analysis.
5. **Hardware Limitations:** Due to hardware constraints, more advanced data imputation techniques could not be implemented, which may have affected the handling of missing or inconsistent data points.

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